Multimodal Nowcasting of Sparse Charge-Exchange Spectroscopy on KSTAR: Fast Diagnostics Recover Ion Temperature Beyond Future-Using Interpolation, but Not Toroidal Rotation

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Abstract

Charge-exchange spectroscopy (CES) is the primary measurement of ion temperature (T_i) and toroidal rotation (V_{rot}) in tokamaks, but photon-collection constraints make it slow and intermittently missing: on the KSTAR 10 ms grid studied here, 8.2% of T_i and 23.9% of V_{rot} samples are absent, the two targets go missing independently, and a further 41.1% of V_{rot} rows are instrument holds, leaving 65.0% of the grid without independent V_{rot} information. We ask whether the always-available fast diagnostics—beam-emission spectroscopy (BES), electron-cyclotron-emission imaging (ECEI), and Mirnov coils—together with the irregular past CES history can recover information at missing CES timesteps that *temporal interpolation of CES alone cannot*. Because the fast diagnostics are dense on the 10 ms grid, we frame the problem as *causal full-grid sequence estimation*: a compact (0.36M-parameter) two-branch recurrent nowcaster consumes every grid row as context, handles the sparse targets in a per-target masked loss rather than by dropping rows, and routes the V_{rot} branch away from the fast diagnostics by construction; a windowed GRU–attention model with observation-masked pooling is the paired control. Evaluation is deliberately adverse: offline interpolation (linear, monotone-cubic PCHIP, local AR, and a Gaussian process) may see *past and future* CES neighbours, while the model is strictly causal, and a *causal* GP represents the strongest deployable baseline. Under a pre-registered protocol—plateau-minimal history (W=2), hold-free training and scoring, file-level splits, a test set never touched by model selection, a shot-clustered paired bootstrap (10,000 resamples), and *two co-primary populations* that treat the 0.53% of T_i spectral-fit failures (>3 keV) as missing or keep them, with an unqualified claim required to hold in both—the nowcaster beats future-using PCHIP on T_i with skill +0.17 to +0.26 (spike-cut) and +0.23 to +0.32 (spike-inclusive), 95% CIs excluding zero on *four independent held-out splits in both populations*, and beats the causal GP on all eight; V_{rot} clears the same gate on 1/4 and 2/4 splits and is reported as a tie. Two stress tests separate the offline from the deployment claim: reweighting the scored points onto the covariate distribution of the *genuinely missing* ones keeps the advantage over every causal method on all splits and over PCHIP on 2/4 (cut) and 4/4 (inclusive); re-splitting strictly in time across campaigns—which removes the windowed control's advantage over interpolation—leaves the sequence nowcaster ahead of PCHIP and of the causal GP on 4/4 initialisations in both populations. The edge concentrates in high-variability neighbourhoods (peak-stratum T_i skill +0.45 to +0.72, 8/8 PASS); calibrated conformal intervals beat both baselines' in every cell; a 21k-parameter latent-bottleneck model with an exact persistence-plus-correction decomposition matches the backbone's T_i skill under the spike cut, and widening the backbone from 34k to 879k parameters changes it by less than ±0.01—the information, not the estimator, is the ceiling of the aggregate loss (46.6% of the squared error sits in the top 1% of one-step changes, so that statement is about a tail statistic). In the median row, where the backbone only equals persistence, a pre-registered hold-or-jump filter with Student-t measurement noise at the training bulk scale sits at 0.91× persistence for T_i on all eight split–population cells and beats the backbone's median for V_{rot} too; because every output that must also win the aggregate gives that

median back, median and mean are reported from separate estimators. Moving context and structure separately separates their roles: stretching contiguous causal context from 20 to 630 ms raises skill from +0.055 to +0.138 and lifts the fraction of discharges beating the causal GP from 0.52 to 0.66 before it flattens, whereas at matched reach recurrence, dilated convolution and attention tie within 0.023—so the architecture is chosen on cost (dispatched operator count), not on skill. Input ablations show the $T_i \leftrightarrow V_{\mathrm{rot}}$ asymmetry is physical: history alone falls below interpolation for T_i in the cut population, while zeroing the fast diagnostics leaves V_{rot} bit-identical (unobserved NBI torque; 100 Hz-sampled Mirnov signals alias away kHz mode rotation). We release the full pipeline, pre-registration, and evaluation artifacts.

1 Introduction

Ion temperature T_i and toroidal rotation V_{rot} at the plasma edge govern key elements of tokamak performance and stability, including pedestal structure, ELM behavior, and access to high-confinement regimes. The workhorse measurement of both quantities is charge-exchange spectroscopy (CES) [1], which infers T_i from Doppler broadening and V_{rot} from the Doppler shift of charge-exchange line emission. Because CES must integrate photons to reach adequate signal-to-noise, it is intrinsically slow, and individual time points are frequently lost to exposure and signal-quality problems. In the KSTAR data studied here, even on a uniform 10 ms grid, 8.2% of T_i and 23.9% of V_{rot} observations are missing—*independently* of each other—while a further 41.1% of V_{rot} rows repeat the previous reading bit-for-bit (§3.4), so only 35.0% of the grid carries an independent V_{rot} measurement. Empirically in our dataset, the missingness concentrates at physically interesting moments (low signal-to-noise phases, ELMs, transitions), i.e. it is missing-not-at-random in the sense of Little and Rubin [2] (MNAR).

Meanwhile, several fast diagnostics observe the same plasma without gaps: BES measures density-fluctuation structure, ECEI images electron temperature, and Mirnov coils record magnetic fluctuations. None of them measures T_i or V_{rot} directly. The question this paper asks is:

> *At a 10 ms timestep where CES is missing, can the simultaneous fast diagnostics plus the irregular past CES history recover T_i and V_{rot} information that temporal interpolation of CES alone cannot?*

A positive answer would justify a data-driven *virtual sensor*: a model that fills CES gaps online, without the strong inverse-mapping assumptions required by hardware alternatives, enabling gap-free pedestal analysis and, ultimately, real-time use.

A deliberately hard bar. The obvious baseline for gap-filling is interpolation of the CES channel itself. We make the comparison intentionally adverse to the model: the interpolation baselines (linear, monotone-cubic PCHIP [3], a local AR reference, and a Gaussian process) are *offline*—they may use CES neighbours on *both sides* of the target time, including the future—whereas the model is strictly causal, seeing only the fast diagnostics up to the target step and the past CES history. If a causal model beats a future-using interpolant, the fast diagnostics must carry CES-relevant information that the temporal structure of CES alone does not contain; and a model that beats future-using interpolation a fortiori beats every causal baseline. Because the strongest *deployable* competitor is not persistence but a past-only smoother, we also add a *causal* GP arm and report the model against it separately.

Contributions.

1. **A causal full-grid framing for sparse-target nowcasting** (§4). The fast diagnostics are dense on the grid, so instead of assembling per-sample windows and dropping unlabelled rows we keep every input-complete row as sequence context and move the sparsity into the loss: a two-branch causal LSTM reads the whole past of a discharge segment with per-target carried-value, staleness and observation-flag channels, and the V_{rot} branch is *structurally* blocked from the fast diagnostics. Against a paired windowed GRU–attention control with observation-masked pooling, this reframing is worth +0.081 pooled T_i skill (16 runs, run-cluster CI $[+0.067, +0.096]$, 16/16 positive) at a fraction of the training cost—and it is the reach it buys that clears the causal GP.

2. **A pre-registered, two-population, shot-clustered evaluation protocol** (§5). History length is set by a plateau-minimal rule from a 24-run sweep (W=2); instrument holds are removed from training and scoring; the 0.53% of T_i spectral-fit failures (>3 keV) define two co-primary populations—treated as missing in every arm, or kept everywhere—and *an unqualified claim must hold in both*. The test set is never read by model selection, significance is a shot-clustered paired bootstrap, and every model decision was fixed in writing before the corresponding TEST scoring.

3. **A robust positive result for** T_i (§6): the causal nowcaster beats future-using PCHIP with $\text{skill}_{\text{PCHIP}} = +0.17$ to $+0.26$ (cut) and $+0.23$ to $+0.32$ (inclusive), CIs excluding zero on all four held-out splits in both populations, and beats the causal GP on all eight split–population cells; it ties the offline GP, the strongest future-using smoother.

4. **A physics-grounded negative result for** V_{rot}, reported as a finding rather than a failure: V_{rot} ties future-using interpolation (1/4 and 2/4 significant, what noise alone would give) while beating persistence on most splits; zeroing the fast diagnostics leaves the V_{rot} output bit-identical and zeroing history collapses it, so rotation remains history-autocorrelation-limited at 10 ms (unobserved NBI torque; Mirnov aliasing). We call this the $T_i \leftrightarrow V_{\text{rot}}$ *information asymmetry*.

5. **Two stress tests that separate two claims usually conflated** (§6.5, §6.6). Reweighting the scored points onto the covariate distribution of the genuinely missing ones keeps the T_i advantage over every causal method on 4/4 splits in both populations and over PCHIP on 2/4 (cut) and 4/4 (inclusive). Re-splitting strictly in time across campaigns removes the windowed control's advantage over interpolation (2/4 and 0/4)—the fast diagnostics drift $1.22\,\sigma$ (BES) between periods against $0.115\,\sigma$ for the targets—while the sequence nowcaster, which standardizes each discharge's inputs by its own statistics, stays ahead of PCHIP and of the causal GP on 4/4 initialisations in both populations.

6. **Where the edge lives**: in high-variability neighbourhoods (peak-stratum T_i skill $+0.45$ to $+0.72$, PASS on 8/8 population–split cells; V_{rot} $+0.54$ to $+0.79$ point estimates, nil in the bulk) and into the non-adjacent regime ($\Delta t > 15$ ms pooled, PASS in both populations).

7. **The complexity ladder and the size axis, both measured** (§6.9): a 21,498-parameter model that predicts *persistence plus a bounded correction read linearly from eight probeable latents* matches the backbone's T_i skill under the spike cut (paired mean $+0.002$)—and loses -0.19 without it, because a bounded correction cannot recover a spiked anchor—while widening the backbone from 34k to 879k parameters moves T_i skill by less than ± 0.01. The input set, not the estimator, is the ceiling of the aggregate loss; the levers that remain are named (§9).

8. **Where the aggregate skill comes from, and the one model-side lever that moves the rest** (§6.11, §6.14): the T_i win is a tail effect—in the median row the nowcaster equals persistence and trails future-using PCHIP by 20%, and the V_{rot} null is a bulk loss, not a tail artifact. A Huber loss whose scale is pinned to the target's own 10 ms reproducibility bound lowers the median error 8–10% at no aggregate cost (cut population). A shot-encoding neural process, a transformer pretrained on another tokamak, and a 4-init ensemble all tie; with no same-shot label, no learned arm beats the training mean on V_{rot}.

9. **A physics filter owns the median** (§6.12): a pre-registered hold-or-jump filter with Student-t noise at the TRAIN bulk scale sits at $0.91\times$ persistence for T_i on 8/8 split–population cells and beats the backbone's median for both targets on 4/4 splits; three ways of folding it into one MSE-winning output all give the median back, so median and mean are reported from separate estimators.

10. **Context and structure, separated** (§6.16): stretching contiguous causal context from 20 to 630 ms raises T_i skill from $+0.055$ to $+0.138$ and the fraction of discharges beating the causal GP from 0.52 to 0.66, while at matched reach a $110\times$ parameter range across recurrence, dilated convolution and attention ties within 0.023. Context makes the win typical; the operator family sets the price, and the price is dispatched operator count.

11. **Deployability, measured** (§8): distribution-free conformal intervals that beat both baselines' intervals in all 32 population–target–variant–split cells without touching the predictor, and stateful one-step inference of the recurrent nowcaster inside the 10 ms cadence on a CPU.

2 Related work

Profile fitting and gap-filling in fusion diagnostics. Gaussian-process regression [4] is the community-standard tool for fitting and uncertainty-quantifying fusion profile data [5, 6] — applied to KSTAR's own CES T_i profiles with SVM-based outlier rejection [7] — and monotone interpolants such as PCHIP [3, 8] are preferred over natural cubic splines [9] near transients because splines exhibit Gibbs-like ringing at ELM- and sawtooth-like steps [10]. Irregularly sampled series admit exact AR-type null models (IAR/CIAR) [11, 12]. We use this ladder—persistence, linear interpolation [13], PCHIP, local AR—as pre-registered baselines, with PCHIP as the headline comparator, and Murphy's MSE skill score [14] as the metric. Our contribution is orthogonal to this literature: rather than a better CES-only interpolant, we test whether *other diagnostics* add information beyond any CES-only method, by benchmarking a causal multimodal model against future-using interpolation.

Neural diagnostic inference in fusion. Neural diagnostic inference has three main lines, and it is instructive to ask of each: *what happens when the target measurement is absent?* The oldest line accelerates spectral fitting—NN analysis of JET CX spectra [15, 16] and fast T_i estimation from EAST CXRS spectra [17], and the newest member, on *our* diagnostic: NN-CES, a physics-constrained network that extracts T_i and rotation from KSTAR CES spectra without beam modulation [18]. These gain orders of magnitude in speed over least-squares fits, but by construction they require measured spectra at the timestep: a missing CES exposure leaves them with no input at all. NN-CES is the closest prior work, so the distinction fits in one sentence: *NN-CES reads the measured timesteps better; this work fills the ones that are not measured.* A second line replaces one inference pipeline with another diagnostic's: Bayesian-model emulation at W7-X [19], profile shapes from scalar parameters on NSTX-U [20], and—closest to our targets—T_i and rotation profiles from an x-ray crystal spectrometer on EAST [21]. Lin et al. predict exactly our two quantities, but from another Doppler spectrometer, i.e. the same physical information channel with the same photon-statistics limits; the mapping is instantaneous and inherits the source diagnostic's availability. A third line builds end-to-end surrogates for slow reconstruction codes: bolometer tomography [22], line-integral diagnostics [23], safety-factor profiles [24], real-time equilibria [25], and—on KSTAR itself—NN Grad–Shafranov solvers [26, 27] and rapid EPED-parameterized profile reconstruction [28] — the latter consuming CES as an *input* and explicitly naming rotation as future work. These address analysis *latency*, not measurement *availability*: they accelerate inversions of data that exist. Two works sit closer to the availability question. Actuator-conditioned profile forecasting on DIII-D [29] is causal and history-driven like our model, but forecasts from *dense* measured profiles for control, rather than recovering a sparse channel. RTCAKENN [30] reconstructs kinetic profiles (including impurity temperature and rotation) in real time and keeps operating when Thomson/CER inputs drop out, via dropout training; missingness there is a robustness condition handled by zeroing inputs, whereas in our setting the target's own sparsity is the problem itself, tracked explicitly through per-target observation flags and irregular-time features. Across all three lines the common assumption is that the information source is available when the model runs; recovering a diagnostic at timesteps where *its own* measurement is missing, from other channels plus its past, falls outside their scope.

Cross-diagnostic reconstruction and temporal densification (closest prior work). A recent family reconstructs one diagnostic from others. GP-based imputation restores faulty magnetic sensors on KSTAR in sub-ms [31]; ML detects and compensates corrupted sensors [32]; autoencoders fuse redundant density diagnostics for resilience [33]; time-series extrinsic regression fills missing electron-temperature data on EAST [34]; and FusionMAE performs masked-channel "virtual backup diagnosis" across 88 channels on HL-3 [35]. Closest to our framing, Diag2Diag synthesizes Thomson-scattering T_e/n_e at fluctuation-diagnostic rates from other simultaneous diagnostics on DIII-D [36], and temporal super-resolution of Thomson profiles from fast radiation diagnostics has been demonstrated on COMPASS [37]; PanoMHD models multimodal signal dynamics with a causal transformer for control [38]. Our work differs from this family on four

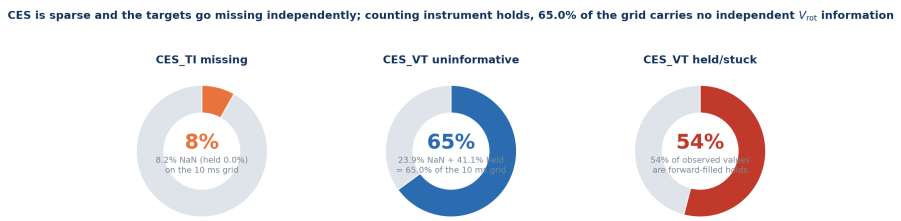


Figure 1: CES missingness structure on the 10 ms grid. T_i and V_{rot} go missing independently (8.2% / 23.9%), motivating per-target masking rather than row filtering; counting instrument holds, 65.0% of the grid carries no independent V_{rot} information.

axes, each verifiable from the cited works' stated scope: (i) *target channel*—every member reconstructs electron quantities or generic dense channels (Thomson T_e/n_e in 36 and 37; ECE-type T_e in 34; arbitrary masked channels in 35), and where CER/CES appears at all it is an *input* [36], never the sparse target; (ii) *causality*—Diag2Diag and the COMPASS study are simultaneous, memory-less mappings, and FusionMAE reconstructs within a window, whereas our model conditions on the target's own irregular past and never reads the future; (iii) *evaluation bar*—none benchmarks against *future-using* interpolation of the target channel, which for gap-filling is the incumbent method to beat, under a pre-registered shot-clustered protocol; and (iv) *information analysis*—reconstructability is assumed rather than tested per target, whereas the $T_i \leftrightarrow V_{\mathrm{rot}}$ asymmetry is our central finding. In sum, this work continues the programme this active family has established—reconstructing an absent measurement from the diagnostics that are always on—and extends it in three directions: from electron channels to a *sparse ion channel*, from simultaneous memory-less mappings to *causal estimation conditioned on the target's own irregular past*, and from assumed reconstructability to *per-target tests against pre-registered baselines*. To our knowledge the conjunction of these three—causal temporal gap-filling of sparse CXRS ion measurements—has not yet been addressed; we present it as the family's natural next step rather than as a departure from it.

Irregular-time and masked multitask learning. Our design borrows standard ingredients from the irregular time-series literature—observation masks and decay-aware recurrence (GRU-D [39]), attention over continuous-time embeddings (mTAN [40]), and latent-ODE dynamics [41]—together with multitask learning under partial labels [42], late-fusion multimodal architectures [43, 44], pre-LayerNorm transformers [45], and attention pooling [46]. The "nowcasting" framing follows its use in machine-learning weather prediction [47, 48]. The novelty is not any single ingredient but the leakage-hardened combination under a physics-adverse evaluation protocol.

3 Data and problem setup

3.1 Dataset

We use 641 KSTAR discharge (shot) files, shots 30801–32751, selected by the data provider for hardware consistency and for H-mode ELM-suppression (RMP) phases, and aligned to a common 10 ms grid: 247,207 rows in total (median 339 per file). Each row provides BES [49] (9 channels), ECEI [50] (4 channels), Mirnov coils (2 channels), time, and the two CES targets T_i and V_{rot} [1]. On this grid T_i is missing in 8.2% of rows and V_{rot} in 23.9%, independently (Fig. 1). Rows are grouped into contiguous segments at time gaps ≥ 0.5 s (a threshold that sits in the valley of a bimodal delta distribution: only 82 of the ∼247k inter-row deltas fall in $(0.1, 0.5)$ s; within a segment 99.4% of steps are 10 ms): a typical file holds **one main segment** (median 301 rows ≈ 3.0 s; 10th–90th percentile 1.3–7.0 s; 28 files have two and 20 have none longer than 10 rows) plus a few isolated single rows (1,279 across the set, typically the first row at $t = 0$) separated from it by gaps of median 6.3 s. No arm—neither an interpolant nor a model's input—bridges a gap.

3.2 Two framings of a training example

The *windowed* framing (our control, §4) builds a sample at target index t from the W preceding grid rows: `bes` $(W, 9)$, `ecei` $(W, 4)$, `mc` $(W, 2)$, `time_features` $(W, 4)$, `ces_history` $(W, 4)$, the normalized target $[T_i, V_{\mathrm{rot}}] \in \mathbb{R}^2$, and a per-target observation mask $m \in \{0, 1\}^2$. The four time features—lookback seconds, inter-row delta seconds, and their $\log(1{+}x)$ transforms—expose the irregular observation pattern explicitly (the reliability of a past CES value depends strongly on whether it is 10 ms or 200 ms old); the four CES-history channels are the previous normalized T_i, previous normalized V_{rot}, and one observation flag per target. $W{=}2$ throughout (§6.8).

The *full-grid sequence* framing (our main model) keeps every input-complete row of a contiguous segment—labelled or not—as context, because the fast diagnostics have no missing values. Each step carries 22 strictly causal channels: the 15 z-scored fast channels, $\log(1{+}\Delta t_{\mathrm{prev}})$, and per target the last observed value carried forward from rows $< t$, $\log(1{+}$staleness$)$, and a has-any-past-observation flag. Targets are z-scored with a parallel observed mask, and the sparsity is handled entirely in the loss. Nothing in either framing lets a model read the target row's own value.

3.3 Design principles

No fabricated labels. We never impute targets to create training rows. In the windowed framing a row is kept iff its diagnostic inputs are complete and *at least one* CES target is genuinely observed; in the sequence framing unlabelled rows contribute context only.

Per-target masked loss. The loss is masked per target, $\mathcal{L} = \sum_{k \in \{T_i, V_{\mathrm{rot}}\}} m_k \, (\hat{y}_k - y_k)^2 / \sum_k m_k$, so a row with only one observed target still supervises that target. A both-targets-required filter would silently discard ≈28% of labelled rows; removing it is a pure data gain.

Leakage hardening. (i) *File-level splits*: adjacent rows within a shot are strongly autocorrelated, so splits are made at the shot-file level and pinned to disk; (ii) *train-file-only normalization*: all z-score statistics (NaN-aware for the sparse targets) are estimated on training files only—the sequence model additionally standardizes each discharge's fast diagnostics by that discharge's own mean and variance, which uses no other discharge and no target (§6.6 shows why this matters); (iii) *target-step masking*: the target row's own values and observation flags never enter the input.

3.4 Data-quality audit 1: held (forward-filled) V_{rot} values

54% of observed V_{rot} values are instrument holds: bit-identical repeats of the previous observation within a contiguous time block (runs up to 1,214 rows; 499/641 files affected). These arise because the native V_{rot} cadence is slower than the row cadence; they are not independent measurements. V_{rot} is recorded to five decimal places with a minimum spacing of 4×10^{-5} between distinct values, so "consecutive-equal" has no measurable false-positive channel; T_i is unaffected (1 held row in 226,991). Under the confirmed protocol holds are removed *everywhere*—from the supervision targets, from the history/carried inputs and their observation flags, from the normalization statistics, and from the interpolation anchors of every baseline—so no arm can score a forward-fill for free. Every number in this paper is on genuine measurements only; the ledger of the project records the earlier hold-inclusive convention and the paired retrain that showed holds also contaminate *training* (a forward-filled staircase teaches a model that copying history is near-optimal), which is why hold-free is the protocol rather than a sensitivity row.

3.5 Data-quality audit 2: T_i spectral-fit failures and the two populations

Observed T_i has p99 = 2,089 eV, p99.9 = 9,601 eV and a maximum of 14,984 eV: the far tail is not plasma physics but failed spectral fits. **1,197 rows (0.53%) exceed 3 keV**, forming 951 runs in 274 discharges; 85% of the runs are single rows, only 2% last five rows or more, and the median run peak is 13× the mean of its observed neighbours (IQR 6–26×). Such rows are unpredictable for every method, and—worse for a fair comparison—a spike *used as an interpolation anchor* poisons the interpolant at its neighbours. Two responses are defensible and each is open to a criticism: treating >3 keV as missing in every arm removes rows no method can predict but invites "the hard rows were removed", while keeping them handicaps the

offline baseline by feeding it spiked anchors. We therefore pre-registered **two co-primary populations**: *cut* (values >3 keV treated as missing at load time, identically in supervision, history inputs, normalization statistics and every baseline's anchors) and *inclusive* (no cut). **A claim is stated without qualification only if it holds in both**; a result that holds in one is reported with its population named. The threshold is immaterial across 2.5–4 keV (§6.15); the value cut is a physically motivated proxy (">3 keV is not a KSTAR ion temperature") pending CES fit-quality metadata, and it is one-sided—the same audit finds single-sample *downward* dips (4,965 rows $\leq \frac{1}{2}$ both neighbours) that no value cut touches, and the cut removes only 19% of the $\geq 2\times$ upward outliers. V_{rot} has fit-failure spikes too (119 rows above 1,000 km/s in 16 shots, 101 of them one block in a single discharge); the protocol leaves V_{rot} uncut and every persistence-anchored V_{rot} comparison reports the share of squared error those rows carry.

4 Model

The main model, `seq_v2`, is a **357,570-parameter** causal sequence nowcaster; the windowed model it replaces is a 201,258-parameter late-fusion network and is retained as the paired control. Before describing either, this section fixes three things: *what is being estimated*, *why those inputs are wired to that target*, and *which time scales the model sits on*. None of the three is measured by the results section; they are what makes those measurements interpretable, and for the routing they make it a consequence of a conservation law rather than a modelling preference.

4.1 The estimand, the loss, and an irreducible floor

At each grid time t the model estimates the normalized target $y_t = [T_i, V_{\mathrm{rot}}]_t$ from the observation history $x_{\leq t}$. The loss is a masked mean squared error over per-target observation masks $m_{t,c} \in \{0, 1\}$, plus a physical penalty discouraging T_i predictions below the normalized zero z_0:

$$\mathcal{L} = \frac{\sum_t \sum_{c \in \{T_i, V_{\mathrm{rot}}\}} m_{t,c} \left(\hat{y}_{t,c} - y_{t,c}\right)^2}{\sum_t \sum_c m_{t,c}} \; + \; \lambda \, \overline{\mathrm{ReLU}\left(z_0 - \hat{y}_{t,T_i}\right)}, \qquad \lambda = 0.1. \tag{1}$$

The minimizer of the first term of (1) is the conditional expectation $\mathbb{E}[y_t \mid x_{\leq t}]$, so what this work reports is the mean of the conditional distribution and not the distribution itself. Uncertainty is handled separately, by distribution-free intervals that leave the model untouched (§8).

This objective has a floor no estimator can remove. The target is the output of a photon-integrated spectral fit, so it carries its own measurement noise σ_{meas}, and even a predictor that recovers the true physical quantity exactly would score

$$\mathrm{RMSE}^2 \; = \; \sigma_{\mathrm{meas}}^2 \; + \; b^2 \tag{2}$$

against it, where b is the physical error. Difference-based estimators applied to the observed series put σ_{meas} for T_i in the cut population at **46.4–129.9 eV** (fourth-difference bound 129.9, trimmed at the 95th percentile 68.7, robust MAD 46.4). The backbone's RMSE of 157.8 eV is the same order as that range, and the reason its two ends differ by nearly a factor of three is what decides the character of the loss: **46.6% of the fourth-difference squared mass (65.6% for V_{rot}) sits in the top 1% of one-step changes.** The aggregate in (1) is therefore a *tail statistic*, and the later finding that width and operator family are flat (§6.9) should be read as a consequence of that fact rather than as an exhaustion of information: in the bulk the model is still 2.3–3.4× above the target's own scatter.

$\lambda = 0.1$ is inherited from the windowed pipeline rather than derived here, and the way to turn it into a derivation is to fix a target violation rate and solve for the multiplier as a constrained optimization. This version instead reports only that the penalty does not manufacture the result: every arm on the ladder is trained with the same λ, and the interpretable rung `b3k8`, which starts at persistence by construction and therefore almost never activates the penalty, matches the backbone (§6.9).

Table 1: Term-by-term audit of (4). Full 641-shot census, confirmed data treatment (spikes cut before held removal), 624 contiguous blocks. r is the lag-0 pooled Pearson correlation between a modality's standardized channel mean and the target, or its one-step increment.

Term	Physics	In our data?	Disposition
$\partial L/\partial t$	rotation rate	difference of consecutive observed pairs	measured
T_{NBI}	beam torque	no 0-D channel exists	absent (§6.7)
$T_{\mathrm{NTV}} \sim \delta B^2$	non-axisymmetric braking	Mirnov, decimated to 100 Hz	closed negative
$\nabla\cdot\Pi_\varphi$	turbulent momentum flux	BES sees the density fluctuation	**measured → null**
T_{int}	residual stress	needs ∇T_i; we hold a scalar	not reachable

4.2 Two conservation laws fix the input routing

The V_{rot} branch of `seq_v2` does not see the fast diagnostics. That is not a modelling preference; it is what reading the two transport equations term by term returns. Ion energy and toroidal angular momentum obey

$$\frac{\partial}{\partial t}\left(\tfrac{3}{2} n_i T_i\right) + \nabla\cdot\mathbf{q}_i = Q_{ei}\,(T_e - T_i) + S_i, \tag{3}$$

$$\frac{\partial}{\partial t}\left(n_i m_i \langle R^2\rangle \omega_\varphi\right) + \nabla\cdot\Pi_\varphi = T_{\mathrm{NBI}} + T_{\mathrm{NTV}} + T_{\mathrm{int}} + \cdots \tag{4}$$

and both constrain a *rate*, not a level. The question to ask about rotation is therefore not whether the fast diagnostics correlate with V_{rot} but whether they correlate with dV_{rot}/dt. Checking each term of (4) against the channels we hold gives Table 1.

T_{NTV} is already closed: Mirnov is decimated to 100 Hz, and amplitude proxies such as |MC| and a rolling RMS never helped (§6.7). The one term left was the turbulent momentum flux, reachable in principle because BES sees the density fluctuation it rides on. The measurement is null:

$$\underbrace{r\big(\mathrm{BES},\, dV_{\mathrm{rot}}/dt\big) = -0.006, \qquad r\big(\mathrm{ECEI},\, dV_{\mathrm{rot}}/dt\big) = -0.003}_{\text{momentum equation}} \quad \text{versus} \quad \underbrace{+0.070, \qquad +0.078}_{\text{energy equation } (dT_i/dt)}$$

The same asymmetry appears at the level of the quantities themselves (BES→T_i +0.341, ECEI→T_i +0.311 against V_{rot} +0.027 and +0.005). *Because the same measurement finds a coupling for the energy equation, the rotation null is not a limitation of the method.* With T_{NBI} absent from the dataset, T_{NTV} closed and T_{int} unreachable, the last term is null as well, so no route exists by which our inputs reach V_{rot} through (4). Splitting the branches at the encoder is the implementation of that audit, and it is why perturbing all 15 fast channels leaves the V_{rot} output bit-identical (§6.7). The remaining lever is acquisition, not modelling (§9).

4.3 The time scales the model sits on

Three times define this problem. First, the grid spacing is $\Delta t = 10$ ms, so the Nyquist frequency is

$$f_{\mathrm{Nyq}} = \frac{1}{2\,\Delta t} = 50 \text{ Hz}. \tag{5}$$

The mode rotation Mirnov would have to carry is in the kHz band—EHO harmonics reported for discharges in this dataset are near 4 and 8 kHz—and a dB/dt stream decimated to 100 Hz without an anti-aliasing filter cannot represent it, by the sampling theorem. The lag-1 autocorrelation of −0.009 against BES's +0.568 (§6.7) is the *confirmation* of that fact, not its basis.

Second, the time that couples electrons to ions. From the Braginskii electron collision time $\tau_e = 3.44\times10^5\, T_e^{3/2}/(Z\, n_e \ln\Lambda)$ (seconds, T_e in eV, n_e in cm^{-3}), the temperature equilibration time is

$$\tau_{\mathrm{eq}} \simeq \frac{m_i}{2m_e}\,\tau_e. \tag{6}$$

For deuterium with $\ln\Lambda = 17$ and $Z = 1$, τ_{eq} spans 8.3–305 ms over $T_e = 0.5$–3 keV and $n_e = 2$–5×10^{19} m^{-3}, and **8–59 ms** in the corner corresponding to our observed T_i median of 593 eV. The saturation point of

contiguous causal context measured in this work is about 50 ms, and the two numbers are the same order. This is an order-of-magnitude consistency statement and nothing more: the plasma parameters in (6) are quoted rather than measured here (BES and ECEI are in instrument units, so this work does not hold n_e and T_e in physical units), and the prefactor carries an order-unity convention.

Third, the time over which a signal resembles itself. Autocorrelation $1/e$ times measured across the 641 files agree to within 10%: T_i 159 ms, BES 161 ms, ECEI 147 ms. V_{rot} does not: 54% of its observations are instrument repeats, so removing held values gives 16 ms and keeping them gives more than 300 ms, a factor of over 19. **The V_{rot} relaxation time cannot be measured in this dataset.** T_i does not suffer this, having one held row in 641 files. The held pathology costs not only 54% of the values but the ability to characterize the quantity's own dynamics, which makes it a third acquisition task alongside NBI torque and the raw kHz Mirnov stream (§9). These autocorrelations are computed without detrending and are dominated by the discharge envelope, so they are not quoted as transport times τ_E or τ_φ.

4.4 Full-grid causal sequence backbone

`seq_v2` runs two independent causal LSTMs over the 22-channel grid sequence of a segment (§3.2). The T_i branch (2 layers, 160 units) reads the full state—fast diagnostics, both targets' carried values, staleness flags and Δt—and the V_{rot} branch (1 layer, 64 units) reads *only the seven non-fast channels*: Δt and the per-target carried value, staleness and observation flag. That wiring is the implementation of the audit in §4.2. Each branch ends in LayerNorm and a small GELU head; outputs are normalized $[T_i, V_{\mathrm{rot}}]$ per row and the loss is (1) applied over every labelled row of the segment. Two consequences follow from the framing. First, the model's *reach* into the past is the whole segment rather than a fixed window—exactly the quantity that separates it from every windowed variant in §6.3. Second, the routing holds at the *encoder*: a shared recurrent state would carry fast-diagnostic information into the V_{rot} head no matter how the head is wired, whereas here perturbing all 15 fast channels leaves the V_{rot} output bit-identical (§6.7).

4.5 The windowed control: observation-masked attention pooling

The control encodes each modality (BES, ECEI, MC) with its own time-aware 1-D CNN over the $W{=}2$ window, reads the CES history with a bidirectional GRU (hidden 64) whose per-timestep outputs are pooled by two independent multi-head additive attention readouts, one per target, *hard-masked* so that attention mass can land only on rows where *that* target was genuinely measured—interpolation's own inductive bias (use only observed samples) written into attention at zero parameter cost. Its T_i head fuses fast diagnostics, time and history; its V_{rot} head fuses history and time only. Everything the two models share—the data contract, hold-free treatment, per-target masking, the split, the scored population—is pinned identically, so their comparison is row-for-row paired.

4.6 The interpretable rung

`b3k8` (§6.9) keeps `seq_v2`'s framing and routing but replaces each head by an *exact* decomposition, $\hat{y} =$ carried value $+ \sum_k w_k z_k + b$, where $z \in [-1, 1]^K$ is a tanh latent from a small causal GRU (64/32 units, $K{=}8$ for T_i, 4 for V_{rot}) and the readout is zero-initialized so training starts exactly at persistence. It has 21,498 parameters, 6% of the backbone.

4.7 The optimization setup, and the state of its justification

Training uses AdamW ($\eta = 10^{-3}$, weight decay 10^{-4}), batch 16 segments, gradient norm clipping at 1.0, `ReduceLROnPlateau` on validation masked MSE (patience 2, factor 0.5) and early stopping (patience 6, cap 30 epochs; the confirmatory runs stopped at 14–25 epochs).

These values are *inherited* from the windowed pipeline rather than derived, and this work does not hide that. The data and structure axes were all closed by controlled experiments—window size (§6.8), held treatment (§3.4), the two populations (§3.5), complexity and width (§6.9)—but the optimization axis never received one. A single, and indirect, piece of evidence defends the setting: the *budget-equalization* condition of the backbone gate trains both models for a fixed 10 epochs with validation-based checkpoint selection

forbidden, and even there the backbone's paired T_i gain is positive on all four splits (+0.063, +0.033, +0.045, +0.030; §7). The conclusion is therefore not an artifact of the training schedule.

The loss axis has since received one controlled experiment (§6.11): a Huber loss with its scale pinned to the target's 10 ms reproducibility bound cut the median error by 8–10% in the cut population while holding the aggregate skill, and the backbone's loss stays MSE by the pre-registered rule. That experiment also taught a coupling: selecting an arm that changes the objective on val MSE picks a nearly untrained checkpoint (epoch 2–4) in the tail-dominated inclusive population. The objective and the checkpoint-selection metric are a pair, and when one is changed, how the other was set is part of the result.

"Why batch 16" now has a measured answer. The gradient noise scale $B_{\text{simple}} = \operatorname{tr}\Sigma/\|G\|^2$ is the batch size at which a gradient's signal and its noise are comparable, and taking the sampling unit to be the block—what the training loop actually draws—it can be computed exactly on the training set rather than estimated. With g_i the gradient of block i's own masked loss,

$$G = \tfrac{1}{N}\sum_i g_i, \qquad \operatorname{tr}\Sigma = \tfrac{1}{N}\sum_i \|g_i\|^2 - \|G\|^2, \qquad B_{\text{simple}} = \frac{\operatorname{tr}\Sigma}{\|G\|^2}. \tag{7}$$

Evaluated over all 452 training blocks at seven points during training, the post-initial median is **33.6 blocks** (geometric mean 28.4, range 6.0–61.4). **Batch 16 sits at about half the measured noise scale.** Below B_{simple} batch size and step count trade roughly linearly, so 16 is neither wasteful nor obviously suboptimal; doubling it to 32 would roughly halve the step count at the same data efficiency, and going far above 60 would begin to waste examples.

This is an order-of-magnitude statement. $\frac{1}{N}\sum_i\|g_i\|^2$ and $\operatorname{tr}\Sigma$ fall monotonically after the first epoch (10.5 → 7.5 and 10.3 → 7.3), but $\|G\|^2$ moves between 0.15 and 1.74, and since it is computed exactly over all 452 blocks that motion is a real property of where the Adam trajectory sits at an epoch boundary rather than measurement error. What is stable is the noise share: after the first epoch **86–98% of a single block's gradient energy is noise**, leaving 2–14% of signal. The measurement is made with dropout disabled, so as to capture the noise from sampling examples alone; the training loop's own effective noise scale is therefore *at least* the figure above, and the figure concerns the compute-versus-steps trade rather than final skill.

The second measurement, μP—a parameterization under which the optimal learning rate does not shift with width—has not been run. Adopting it would make the distinction between "capacity does not help" and "the learning rate did not match that width" unnecessary by construction for the width sweep of §6.9.

What the search taught. Across two rounds of controlled architecture iterations on the windowed family (selection on clean validation skill only), attention pooling was the one mechanism that survived, and capacity scaling, skip topologies and extra extractors never helped. Under the confirmed protocol the same lesson recurs twice more: observation-masked causal attention added to `seq_v2` is a small, consistently positive but non-significant gain (§7), and a five-point width sweep of `seq_v2` is flat (§6.9). The bottleneck in this problem is information, not capacity, and (2) and Table 1 say where that bottleneck sits for each target.

5 Evaluation methodology

Metric. All errors are denormalized to physical CES units and computed per target. The primary score is Murphy's skill [14] against the pre-registered headline baseline, $\text{skill}_{\text{PCHIP}} = 1 - \text{MSE}_{\text{model}}/\text{MSE}_{\text{PCHIP}}$ (positive = model better; 0 = tie). Every arm (model and all baselines) is scored on the *identical* (file, row) sample set under the same per-target keep mask, verified bit-identically on the population keys before any paired comparison. Interpolants exclude the target's own value (no leakage). Neither the interpolants nor a model's input bridges a segment boundary; where a neighbour beyond the boundary would be needed the interpolant predicts the persistence value instead (no population shrinkage; PR2).

Baseline ladder. Persistence and a local past-only AR are the causal references of the original pre-registration; linear, PCHIP and a Gaussian process (exact Matérn-3/2 + white noise, local nearest-16+16 fit, per-sample marginal-likelihood grid selection, inheriting the PR2 fallback) read the future. To the causal side we add a **causal GP**: the same GP restricted to the 16 past neighbours, with the same NaN conditions so the scored population does not move. It is decisively the strongest deployable baseline (seed-42 T_i RMSE 164.3

against persistence's 197.2, cut population), and "beats every deployable causal method" is judged against it, not against persistence.

Three-way split; selection never sees TEST. Files are split train/val/test; the test files of every split were reserved *before* any architecture search and are never read during model selection. The canonical (seed-42) test set contains $n = 32{,}589$ observed T_i rows across 96 shots and $n = 10{,}463$ genuine $V_{\rm rot}$ rows across 60 shots in the cut population (32,721 / 10,461 inclusive); the four splits (seeds 42, 1, 7, 123) hold 32.6–35.9k T_i rows over 96 shots and 10.5–14.5k $V_{\rm rot}$ rows over 60–66 shots each. Every number in this paper is read from frozen run directories by a single collection script, so the text, the tables and the figures cannot drift apart.

Pre-registration. Rules fixed in writing before the corresponding test numbers were viewed: **PR1** the headline comparator is PCHIP (chosen a priori for monotonicity/no ringing at transients; the full ladder is also reported); **PR2** interpolation predicts everywhere the model is scored, with persistence fallback where no future neighbour exists—and the fallback rate is reported (0.3–0.4% of scored T_i rows, but **40–44% of scored $V_{\rm rot}$ rows**, so "vs. PCHIP" for $V_{\rm rot}$ is two-fifths "vs. persistence"); **PR3** a minimum test-set floor ($\geq$15 shots, $\geq$3,000 observed T_i samples; met everywhere); **PR4** significance means a shot-clustered bootstrap 95% CI on the skill excluding zero. To these the confirmed protocol adds: hold-free training and scoring; $W{=}2$ by the plateau-minimal rule of a 24-run sweep (§6.8); a per-file sample cap of 500 in the windowed family; the two co-primary populations of §3.5; and *model-decision rules committed before TEST scoring* for the backbone gate, the one attention candidate, the interpretable rung and the width sweep (§7). Threshold sensitivity (2.5/3/4 keV) is reported.

Shot-clustered paired bootstrap. Adjacent CES rows within a discharge are strongly correlated; treating samples as independent would grossly understate uncertainty. Per-sample paired errors $\mathrm{SE}_{\rm model} - \mathrm{SE}_{\rm PCHIP}$ are aggregated by shot, and whole shots are resampled with replacement (10,000 resamples, fixed seed). The resulting CI reflects genuine shot-to-shot generalization; its cost is that the effective sample size is the number of shots ($\approx$96 for T_i, 60–66 for $V_{\rm rot}$), which bounds statistical power throughout. Model-vs-model comparisons (backbone vs. control, rung vs. backbone) use the same paired bootstrap on the same rows.

The MNAR problem, and what we do about it. Truly missing points have no ground truth, so skill is measured by masking *observed* points. Because CES missingness is MNAR (drop-out at low S/N, ELMs, transitions), observed-point skill is an optimistic estimate of performance at genuinely missing points. §6.5 reweights the scored points onto the covariate distribution of the *actually missing* ones and reports how much of the skill survives, together with what fraction of the missing population is in the analysis' domain at all.

6 Results

6.1 The nowcaster dominates every causal baseline

Table 2 shows the physical-unit RMSE ladder on the canonical test split. `seq_v2` has the lowest RMSE for both targets among all methods that do not read the future, beats the causal GP—the strongest deployable arm—by 4% (T_i) and 18% ($V_{\rm rot}$) in RMSE, and edges the future-using interpolants; the offline GP, which averages noisy anchors rather than passing through them, is the one arm it ties (T_i 153.8 vs. 157.8). This is the most robust result in the paper: for any online setting, where future CES is by definition unavailable, the sequence nowcaster is the clear winner.

6.2 Headline: T_i beats future-using interpolation on four independent splits, in both populations

Table 3 reports the backbone on the four independent held-out splits (seeds 42, 1, 7, 123—the latter three never used in any selection step) in the two co-primary populations, against PCHIP and against the causal GP; Fig. 3 shows the PCHIP intervals.

Table 2: Physical-unit RMSE ladder, canonical held-out test split (seed 42), cut population, genuine measurements only (n = 32,589 T_i, 10,463 V_{rot}). Lower is better. The three light rows read the future. Inclusive population: `seq_v2` 363.0 / 23.7, PCHIP 412.4 / 30.2, causal GP 394.6 / 28.8, persistence 478.0 / 33.4 (T_i / V_{rot}); the fit-failure spikes more than double every T_i RMSE without changing the order.

Arm	Information access	T_i RMSE (eV)	V_{rot} RMSE (km/s)
`seq_v2` (nowcaster)	fast diag. + past CES, whole segment	**157.8**	**23.6**
Windowed control (W=2)	fast diag. + 2 rows of past CES	169.2	26.1
Causal GP	past 16 CES neighbours	164.3	28.8
Persistence	last observed CES	197.2	33.4
AR (local, causal)	past CES only	472.2	51.0
GP (offline)	past + future CES	153.8	24.7
Linear interpolation	past + future CES	169.8	29.0
PCHIP interpolation	past + future CES	173.6	30.2

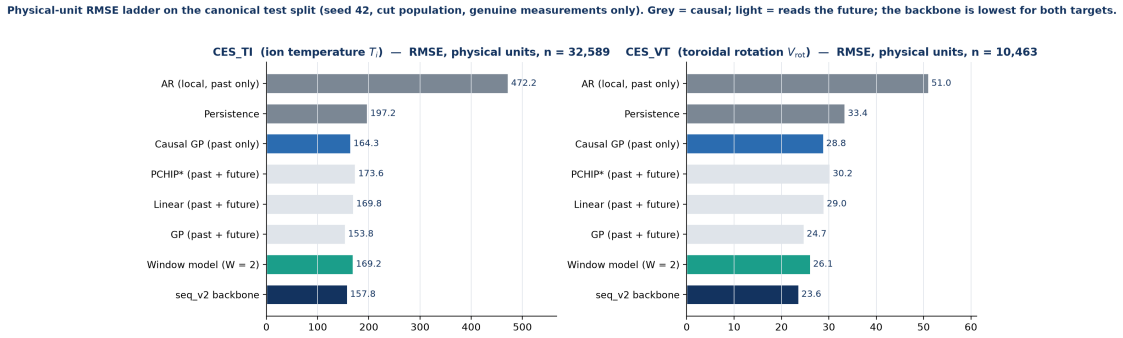


Figure 2: RMSE ladder for the seed-42 test split (cut population, genuine measurements): the causal nowcaster attains the lowest RMSE among deployable methods for both targets and edges the future-using interpolants. Same numbers as Table 2; both are generated from the same frozen artifacts.

- **T_i: PASS on all four splits, in both populations.** $\mathrm{skill}_{\mathrm{PCHIP}}$ = +0.17 to +0.26 (cut) and +0.23 to +0.32 (inclusive); every shot-clustered CI excludes zero, and every one of the eight population–split cells also beats the *causal GP* (+0.08 to +0.17) and persistence (+0.36 to +0.46). This is the unqualified claim of the paper. Against the offline GP the backbone *ties* (point estimates −0.05 to +0.11, significant on 1/8 cells), so the observed-population claim reads "beats the pre-registered interpolants and every deployable causal method, and matches a future-using, per-sample-tuned smoother".
- **V_{rot}: a tie, not a win.** Point estimates are positive on every split in both populations, but PR4 passes on 1/4 (cut) and 2/4 (inclusive)—seed 42 in both, seed 1 at $[+0.000, +0.286]$ inclusive. Against persistence V_{rot} passes on 3/4 splits in both populations (+0.30 to +0.50; seed 7's CI touches zero) and against the causal GP on 2/4. One or two significant splits out of four is what noise would produce, so we claim no rotation win.
- **Why the inclusive numbers are higher.** Every arm scores *better* against PCHIP without the cut (backbone mean +0.268 vs. +0.236; control +0.238 vs. +0.173), because the spikes poison the interpolation anchors: a learned model discounts a spiked carried value where an interpolant cannot. §6.7 quantifies this component; it is the reason neither population is quoted alone.

6.3 The full-grid framing against the windowed control: the backbone gate

Was the sequence framing worth adopting? The decision was pre-registered as a gate with four conditions and scored once (cut population, the population under which internal model selection is done): a 4×4 grid of split seed × initialisation seed, each run paired row-for-row against its split's W=2 windowed control. Paired

Table 3: Held-out test $\text{skill}_{\text{PCHIP}}$ of `seq_v2` with shot-clustered 95% CIs (10,000 resamples), both populations; the last column is the paired skill against the causal GP on the same rows. Bold = PASS (CI excludes 0).

Target	split	cut: $\text{skill}_{\text{PCHIP}}$ (95% CI)	inclusive: $\text{skill}_{\text{PCHIP}}$ (95% CI)	vs. causal GP (cut / incl.)
T_i	42	**+0.174** [+0.097, +0.236]	**+0.225** [+0.109, +0.293]	**+0.078** / **+0.154**
	1	**+0.248** [+0.188, +0.295]	**+0.238** [+0.153, +0.302]	**+0.133** / **+0.169**
	7	**+0.257** [+0.199, +0.302]	**+0.292** [+0.232, +0.344]	**+0.138** / **+0.123**
	123	**+0.264** [+0.188, +0.320]	**+0.316** [+0.186, +0.392]	**+0.105** / **+0.149**
V_{rot}	42	**+0.390** [+0.077, +0.591]	**+0.384** [+0.066, +0.586]	**+0.331** / **+0.324**
	1	+0.183 [−0.028, +0.280]	**+0.195** [+0.000, +0.286]	+0.130 / **+0.143**
	7	+0.135 [−0.358, +0.269]	+0.132 [−0.362, +0.266]	+0.020 / +0.018
	123	+0.305 [−0.049, +0.437]	+0.304 [−0.065, +0.433]	**+0.134** / +0.132

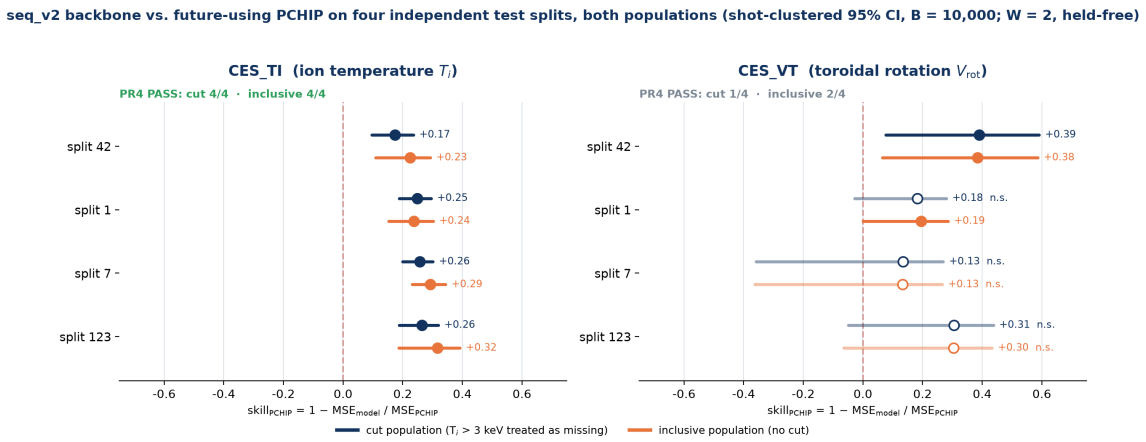


Figure 3: Forest plot of `seq_v2`'s $\text{skill}_{\text{PCHIP}}$ with shot-clustered 95% CIs across four independent held-out splits, both populations. T_i excludes zero on all eight cells; V_{rot} on three.

T_i skill was **positive on 16/16 runs and individually significant on 13/16**; the per-split initialisation means are +0.129, +0.059, +0.078, +0.058 (initialisation spread $\ll$ split spread); the pooled mean is **+0.081** with run-cluster 95% CI [+0.067, +0.096]; a budget-equalized arm (fixed 10 epochs, final weights, no validation selection) keeps the sign on 4/4 splits (+0.063, +0.033, +0.045, +0.030), so the gain is architectural rather than a training-budget effect; and no run showed a significant V_{rot} deficit (8/16 significant wins). All four conditions held. On the four confirmatory splits reported everywhere else in this paper the same comparison reads +0.130, +0.058, +0.062, +0.044 (cut) and +0.053, +0.024, +0.047, +0.029 (inclusive), positive on 8/8, significant on 2/4 in each population.

The gate also settles *what* the framing buys. The windowed control ties the causal GP (significant on 1/4 splits, cut) and the sequence backbone beats it on 4/4 in both populations: the reach into the whole past of a segment is what clears the strongest deployable baseline. Its cost is negative—the backbone trains in a fraction of the windowed family's time because there is no per-sample window assembly or combinatorial augmentation.

6.4 Where the skill lives: gap regime

Stratifying by Δt, the time since the last observed CES value, $\geq$96% of scored targets lie in the adjacent regime ($\Delta t \leq 15$ ms); per split the wide bins hold only hundreds of samples, so we *pool the four test splits* and cluster the bootstrap on the physical discharge (Table 4). As Δt grows PCHIP's task gets *easier*—it interpolates between two real observations that straddle the gap—while ours gets harder, so this is the regime where a causal method is least expected to win.

Table 4: Gap-stratified skill, four test splits pooled, shot-clustered 95% CIs, `seq_v2`. Bold = CI excludes zero. Rows with fewer than 50 samples are not scored.

Δt	n (cut / incl.)	shots	cut: vs. PCHIP	inclusive: vs. PCHIP
T_i				
≤ 15 ms	134,546 / 135,317	301	**+0.239** [+0.197, +0.274]	**+0.299** [+0.244, +0.347]
> 15 ms	3,422 / 3,334	265 / 263	**+0.268** [+0.187, +0.337]	**+0.206** [+0.108, +0.290]
> 45 ms	460 / 429	104 / 101	**+0.267** [+0.092, +0.414]	−0.004 [−0.304, +0.246]
V_{rot}				
≤ 15 ms	51,689	197	**+0.233** [+0.020, +0.318]	**+0.233** [+0.020, +0.317]
> 15 ms	466 / 456	130	**+0.418** [+0.104, +0.680]	**+0.432** [+0.128, +0.696]
> 45 ms	14	7	—	—

The T_i win is not confined to adjacent history. Beyond 15 ms the backbone beats future-using PCHIP in both populations (+0.268 and +0.206, both CIs above zero) and beats persistence there by +0.40 and +0.43. Beyond 45 ms it still wins under the cut (+0.267) and *ties* in the inclusive population (−0.004, CI [−0.30, +0.25]): 429 rows over 101 shots is exactly where a handful of spike-anchored rows can dominate a stratum, and we report the tie as such.

V_{rot} beats interpolation in the one place it is hardest for interpolation. The global V_{rot} tie is a tie in the adjacent regime; over $\Delta t > 15$ ms the backbone beats PCHIP in both populations (+0.418 and +0.432, 130 shots)—the single positive unconditional V_{rot} verdict in the paper—and beats persistence at ≤ 15 ms (+0.39). The widest bin has 14 rows and is not scored.

6.5 How much of this survives at the points that are actually missing?

Everything above is measured on CES points that happen to be *observed*. We can never measure error where there is no ground truth—but we can measure how skill varies over covariates that are computable at observed *and* missing rows, and reweight. Two covariates carry the mechanism: Δt since the last genuine observation, binned at 15/25/45 ms, and the input-only local-activity flag of §6.10, which is defined at a missing row because it is computed from a neighbour set that excludes the row itself. We post-stratify on $\Delta t\times$ activity, drop any stratum with fewer than 30 scored samples, and report the surviving coverage next to every estimate; the weight grid is population-consistent (the cut is applied to it too). The assumption—that within a stratum missing and observed rows are exchangeable—is stated, not assumed away.

First, a scope result. The analysis' domain is "a genuine observation of that target within two rows", the $W{=}2$ criterion. Of the genuinely missing T_i rows, **54–68% are in-domain** per split; of the genuinely missing V_{rot} rows, **only 4–6%** are, because V_{rot} gaps are long blocks (coverage of surviving strata 0.99 for T_i, 0.73–0.76 for V_{rot}). The reweighted V_{rot} row therefore answers a question about one-twentieth of the missing mass and we draw nothing from it.

Second, the two comparisons come apart—and the split is the useful thing. Against **persistence—the baseline an online system actually runs—the T_i advantage survives reweighting on all four splits in both populations** (+0.28 to +0.44, every CI excluding zero); the MNAR correction costs at most 0.12 of skill and on some splits nothing. Against **future-using PCHIP** the reweighted point estimates stay at +0.14 to +0.28 but the wide fixed-weight CIs pass through zero on two cut-population splits (2/4 significant; 4/4 inclusive). Missing points sit at larger Δt, which is where a two-sided anchor helps most and where the reweighting bootstrap is thinnest. The claim the evidence supports at the points that matter is therefore: *at genuinely missing, in-domain timesteps the nowcaster is significantly better than any causal CES-only method in both populations, and better than offline interpolation with a population-conditional significance (2/4 cut, 4/4 inclusive).* The windowed control behaves the same way (2/4 and 4/4 vs. PCHIP; 4/4 vs. persistence).

Table 5: T_i skill of `seq_v2` reweighted from the observed population onto the genuinely-missing, in-domain one (shot-clustered 95% CIs at fixed stratum weights). Bold = CI excludes zero; *unweighted* is the headline of Table 3.

		vs. PCHIP		vs. persistence	
Population	split	unweighted	missing-matched (95% CI)	unweighted	missing-matched (95% CI)
cut	42	+0.174	+0.140 [−0.071, +0.290]	+0.360	**+0.398** [+0.278, +0.518]
	1	+0.248	**+0.164** [+0.062, +0.250]	+0.406	**+0.366** [+0.299, +0.448]
	7	+0.257	+0.203 [−0.024, +0.346]	+0.403	**+0.310** [+0.227, +0.398]
	123	+0.264	**+0.283** [+0.069, +0.392]	+0.415	**+0.381** [+0.246, +0.464]
inclusive	42	+0.225	**+0.140** [+0.033, +0.250]	+0.423	**+0.443** [+0.141, +0.623]
	1	+0.238	**+0.217** [+0.086, +0.319]	+0.436	**+0.383** [+0.273, +0.536]
	7	+0.292	**+0.167** [+0.067, +0.265]	+0.406	**+0.283** [+0.172, +0.380]
	123	+0.316	**+0.221** [+0.050, +0.320]	+0.459	**+0.337** [+0.164, +0.455]

Table 6: Temporal (campaign) split, four initialisations, TEST block of 97 later discharges. Entries are $\text{skill}_{\text{PCHIP}}$; **bold** = PR4 PASS. The last column is the paired T_i skill of `seq_v2` against the windowed control on the same rows.

Population	arm	T_i vs. PCHIP (init 42 / 1 / 7 / 123)	PASS	T_i vs. causal GP	`seq_v2` − control
cut	windowed control (OFF)	+0.027 / **+0.091** / −0.001 / **+0.061**	2/4	0/4	—
	control, per-shot std. (ON)	**+0.103** / **+0.107** / **+0.094** / **+0.107**	4/4	—	—
	`seq_v2`	**+0.187** / **+0.174** / **+0.181** / **+0.177**	4/4	4/4 (+0.11 to +0.12)	**+0.164** / **+0.091** / **+0.182** / **+0.124**
inclusive	windowed control (OFF)	+0.014 / +0.047 / +0.055 / +0.089	0/4	0/4	—
	`seq_v2`	**+0.173** / **+0.202** / **+0.198** / **+0.184**	4/4	4/4 (+0.13 to +0.16)	**+0.161** / **+0.163** / **+0.151** / **+0.104**

6.6 Does it survive a campaign shift?

The second thing deployment does that a random split does not is move forward in time. We order all 641 discharges by shot number (30801–32751) and cut strictly in time: **train 416 shots [30801, 31991], val 128 [32002, 32310], test 97 [32312, 32751]**, no test shot preceding any train shot. Everything else is pinned to the headline protocol, so the only changed variable is the split rule; the split is fixed, so the four runs vary the *initialisation* seed only, and we say so rather than presenting them as four splits. Three arms per population: the windowed control as trained for the headline (train-file-only normalization), the same control with per-shot standardization of the fast diagnostics (cut population), and `seq_v2` (Table 6, Fig. 4).

Three things in that table are worth stating explicitly. First, the *windowed* model's advantage over offline interpolation does not survive the shift: 2/4 (cut) and 0/4 (inclusive) initialisations, and 0/4 against the causal GP. Second, its named repair works: standardizing each discharge's fast diagnostics by *its own* mean and variance at load time (targets untouched) takes the cut-population gate from 2/4 to 4/4, paired +0.078, +0.018, +0.095, +0.049 (3/4 significant), and leaves V_{rot} unchanged (−0.003 to +0.008)—the expected negative control, since the V_{rot} head never sees the fast diagnostics. Third, and this is the result of the section: **the sequence nowcaster does not collapse**. It beats PCHIP on 4/4 initialisations in both populations (+0.17 to +0.20), beats the causal GP on 4/4 in both (+0.11 to +0.16), and beats the windowed control on 8/8; its V_{rot} beats persistence on 4/4 in both populations (+0.31 to +0.34) where the windowed control's does on 0/4, and its V_{rot} margin over the control is significant on 8/8 (+0.06 to +0.18).

Why the control degrades—measured, not asserted. What fails to transfer should be the *fast-diagnostic* pathway, since that is what buys the edge over CES-only interpolation, while the CES-history pathway sees the same physical quantity in the same units in both periods. Measuring the train→test drift per channel, in the units the control was normalized in, confirms it: the median location shift is **1.22 σ for BES** and **0.53 σ for ECEI**, with scale ratios of 0.75 and 0.62—against **0.115 σ and a scale ratio of** 1.06 **for the CES targets themselves**. The fast diagnostics drift 5–11× more than the quantity being predicted. Train-file-only normalization (§3) is the correct leakage-hardening choice under a random split and is precisely what breaks under campaign shift; per-shot standardization uses only that discharge's own data, so it stays leak-free. `seq_v2` standardizes per shot by definition—which is one of the two reasons it transfers; the other

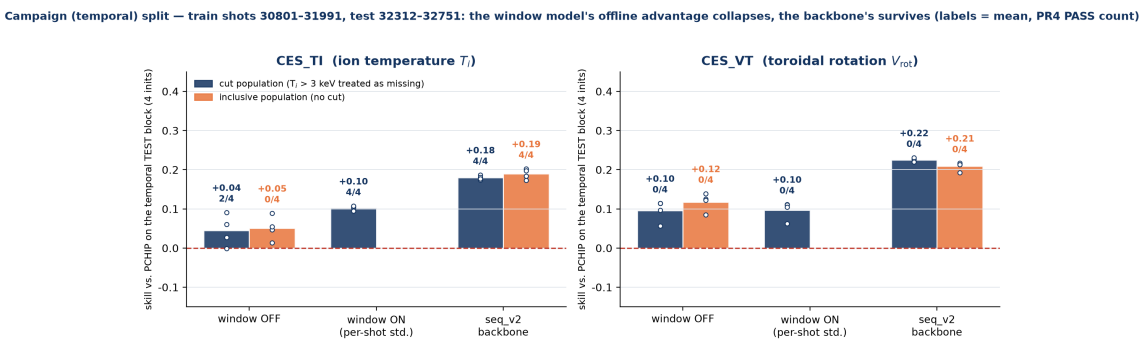


Figure 4: Campaign (temporal) split: the windowed control's advantage over interpolation collapses, per-shot standardization repairs it, and the sequence nowcaster keeps it in both populations. Bars are 4-initialisation means with per-initialisation points and PR4 counts.

Table 7: The two claims under stress, `seq_v2`, T_i. Entries are PR4 pass counts (cut / inclusive) with the point-estimate range.

Evaluation	vs. PCHIP (offline, future-using)	vs. causal baselines
Random split, observed points	**4/4 / 4/4**, +0.17 to +0.32	**4/4 / 4/4** vs. causal GP, +0.08 to +0.17
Reweighted to missing points (§6.5)	2/4 / **4/4**, +0.14 to +0.28	**4/4 / 4/4** vs. persistence, +0.28 to +0.44
Temporal campaign split (§6.6)	**4/4 / 4/4**, +0.17 to +0.20	**4/4 / 4/4** vs. causal GP, +0.11 to +0.16

is reach (§6.3).

Two stress tests, one reading. Table 7 puts them side by side for the adopted model. The advantage over the baselines a real-time system can actually run survives both stress tests in both populations. The advantage over *future-using* interpolation survives the temporal shift in both populations and the MNAR reweighting with population-conditional significance. The version of this table for the windowed model—which the earlier draft of this work reported—had the offline advantage surviving *neither* test; the sequence framing is what changed it, and we say so rather than presenting the new column as if it had always been there.

Caveat that stays. The four campaign passes are initialisations on one temporal test block, so this is one temporal split's evidence, not four; and two of the four cut-population `seq_v2` campaign runs stopped at the 30-epoch cap.

6.7 The $T_i \leftrightarrow V_{\mathrm{rot}}$ information asymmetry

A controlled input-modality ablation on the windowed control—one modality group zeroed *at evaluation*, no retraining, persistence and interpolants computed from the real history—explains *why* the two targets behave differently and re-grounds the routing of §4 under the confirmed protocol (Table 8, Fig. 5).

- **History is indispensable for both targets.** Without the CES history the windowed model falls to −1 to −4 against PCHIP on both targets in both populations: neither target is predictable from 100 Hz diagnostics without its own anchor.
- **T_i: the margin over interpolation is fast-diagnostic information—in the cut population.** With BES/ECEI/MC zeroed the control drops *below* the interpolator (−0.10 to −0.18; paired −0.25 to −0.43, 4/4 significant). The physical channel is collisional electron–ion coupling ($t_{ei} \propto T_e^{3/2}/n_e$): ECEI supplies T_e and BES supplies n_e structure.
- **In the inclusive population the same history-only model still beats PCHIP by** +0.15 **to** +0.23, and the fast channels add only 0.03–0.09 (2/4 significant): the interpolant's anchors are spiked

Table 8: Evaluation-time modality ablation of the W=2 windowed control, TEST, both populations: $\text{skill}_{\text{PCHIP}}$ (4-split mean) and the paired change against the un-ablated run (splits 42 / 1 / 7 / 123; * = significantly worse).

Inputs	target	cut	inclusive
Full (history + fast + time)	T_i	+0.173	+0.238
History + time only (*no fast*)	T_i	−0.125; paired −0.25* / −0.38* / −0.42* / −0.43*	+0.201; paired −0.03* / −0.04 / −0.03 / −0.09*
Fast + time only (*no history*)	T_i	−2.11; paired −4.6* / −1.8* / −2.3* / −1.9*	−1.16; paired −1.5* / −3.4* / −1.2* / −1.1*
Full	V_{rot}	+0.213	+0.206
History + time only (*no fast*)	V_{rot}	+0.213; paired +0.000 ×4 (**bit-identical**)	+0.206; paired +0.000 ×4 (**bit-identical**)
Fast + time only (*no history*)	V_{rot}	−2.89; paired −5.4* / −6.3* / −1.8* / −2.3*	−3.51; paired −6.9* / −7.8* / −1.9* / −2.2*

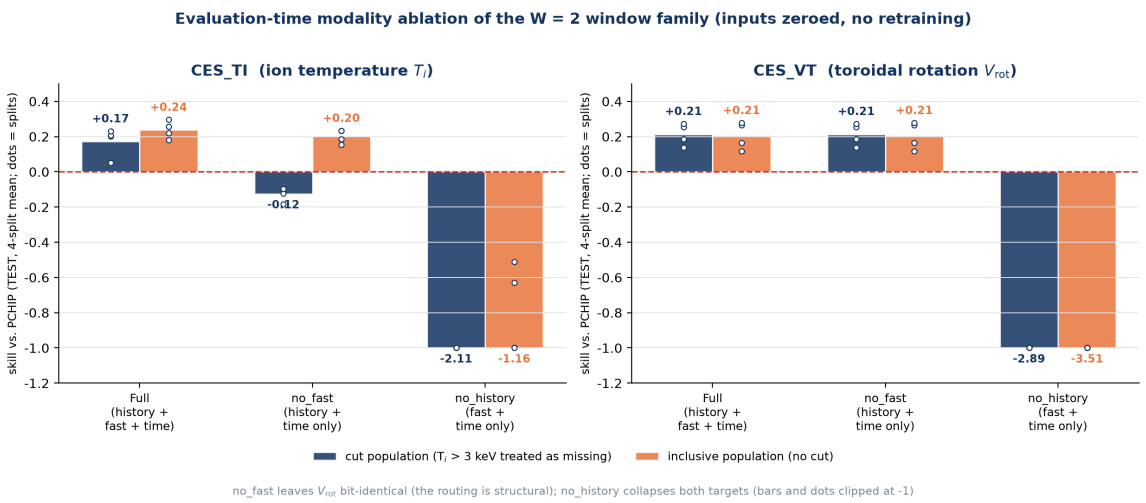


Figure 5: Evaluation-time modality ablation of the windowed control, both populations. Zeroing the fast diagnostics leaves V_{rot} bit-identical and removes the T_i margin under the cut; zeroing history collapses both targets.

and a learned model discounts them where an interpolant cannot. The spike-inclusive margin therefore contains a *spike-robustness* component that is not fast-diagnostic information, which is the measured reason this paper reports two populations rather than the more flattering one alone; the cut population is the one that isolates what the fast diagnostics contribute.

- V_{rot}**: information is entirely CES history, by construction and by measurement.** Zeroing the fast diagnostics leaves the V_{rot} output *bit-identical* on 8/8 population–split cells (the routing is structural), and `seq_v2`'s and `b3k8`'s V_{rot} branches pass the same perturbation test. Toroidal rotation is governed by momentum sources none of our inputs observe—the external NBI torque and intrinsic rotation drives [51]—and the Mirnov signals are sampled at 100 Hz, aliasing away the kHz mode-rotation frequencies that could otherwise proxy rotation (§9).

This asymmetry was predicted from physics before the ablation and is confirmed by it; the V_{rot} non-win is a finding about diagnostic information content, not a model failure. It also motivates the per-target routing of §4.

6.8 How much history is needed? One observation

The history window of the windowed family was originally an unexamined default (W=4). We replaced the assertion with a curve and a stated selection rule: **24 independent runs**, $W \in \{2, 3, 4, 6, 8\}\times$ seeds $\{42, 1, 7, 123\}$, plus a *history-0* point ×4 seeds, each scored on its own held-out test split with the same protocol as the headline (hold-free, per-file cap 500, no spike cut—this sweep predates the cut, and its frozen W=2 runs are the inclusive-population control of §6.2). Three controls make the points comparable: the

Table 9: Window sweep, mean held-out test skill$_{\text{PCHIP}}$ over 4 seeds (PASS = seeds whose shot-clustered 95% CI excludes zero). History-0 is the `no_history` ablation at W=4; W=1 is not constructible.

History obs.	W	T_i skill	PASS	V_{rot} skill	PASS
0	4	−0.026	0/4	−0.783	0/4
1	2	+0.238	**4/4**	**+0.206**	0/4
2	3	**+0.246**	**4/4**	+0.203	1/4
3	4 *(old default)*	+0.221	3/4	+0.190	1/4
5	6	+0.190	3/4	+0.205	1/4
7	8	+0.216	**4/4**	+0.204	2/4

Table 10: Complexity ladder, held-out TEST skill$_{\text{PCHIP}}$ for T_i (4-split mean), and the paired verdicts of the pre-registered rule, both populations. The anchor+Δ model collapses onto persistence at W=2 (its slope term needs two observed history rows and never fires).

Arm	params	cut	inclusive	`b3k8` − anchor (T_i, cut / incl.)	`b3k8` − `seq_v2` (T_i, cut / incl.)
Persistence	0	−0.264	−0.288		
Anchor+Δ (named terms)	1,258	−0.261	−0.287	+0.35 to +0.42, 4/4* / +0.29 to +0.34, 4/4*	mean **+0.002** (all CIs ∋ 0) / mean **−0.194** (4/4*)
`b3k8`	**21,498**	**+0.237**	+0.126		
Windowed control	201,258	+0.173	+0.238		
`seq_v2` backbone	357,570	+0.236	+0.268		

file-level split is window-invariant (all runs evaluate the same 96 test shots per seed); the per-file cap prevents temporal-subset augmentation—which explodes from 240k samples at W=2 to 30.1M at W=8—from letting a few long-block discharges dominate; and the scored population shrinks only 1.8% from W=2 to W=8.

History is essential, and its first observation carries essentially all of it. Removing previous-CES entirely puts T_i *below* PCHIP (−0.026, 0/4) and V_{rot} at −0.78. A *single* past observation lifts both targets to their plateau at once and the curve is flat from there: T_i spans 0.190–0.246 and V_{rot} 0.190–0.206, while the seed spread *within* a single point is 0.07–0.16, wider than the whole curve. Applying "smallest W that reaches plateau" returns $\mathbf{W = 2}$; the incumbent W=4 is below the W=2/W=3 values on both targets, so nothing in the data supports paying for a longer window on accuracy grounds. The one defensible reason to go wider is **coverage**: a windowed sample is scored only when an observed value exists inside the window, so $\Delta t > 15$ ms scored rows go 456 → 1,958 and $\Delta t > 45$ ms 14 → 135 from W=2 to W=8. That is not an argument for a longer window; it is the argument for the sequence framing, whose reach is the whole segment and whose W is not a hyper-parameter.

6.9 What does the complexity buy, and does size help? A ladder and a width sweep

A fair criticism of any learned nowcaster is that its parameters are unexplainable. The useful reply is to *measure what the complexity earns*, under the same protocol, with decision rules fixed before TEST scoring.

The interpretable rung. `b3k8` (§4) predicts $\hat{y} = \text{carried value} + \sum_{k=1}^{K} w_k z_k + b$ with $z \in [-1, 1]^K$ from a small causal GRU and a zero-initialized readout, so (i) the decomposition into K+1 named terms is exact, not an attribution approximation; (ii) every number the readout can see is one of K probeable latents; (iii) the correction is bounded by $\sum_k |w_k| + |b|$ in target σ. K was the one explored variable (val only, two splits; K=8 chosen), and both arms stop by the same patience rule with a non-binding cap. Two pre-registered conditions: beat a retrained named-terms anchor+Δ model (1,258 parameters) on 4/4 splits with no V_{rot} deficit, and stay within −0.05 of the backbone's T_i skill on average.

- **Under the cut, all of the backbone's T_i skill survives compression to eight bounded numbers plus persistence.** `b3k8` beats the retrained anchor on 4/4 splits with no V_{rot} deficit, and its paired T_i skill against the backbone is −0.009, −0.005, +0.026, −0.004 (mean +0.002, every CI containing zero); it passes PR4 vs. PCHIP 4/4 and beats the causal GP 4/4. Linear probes on TEST inputs

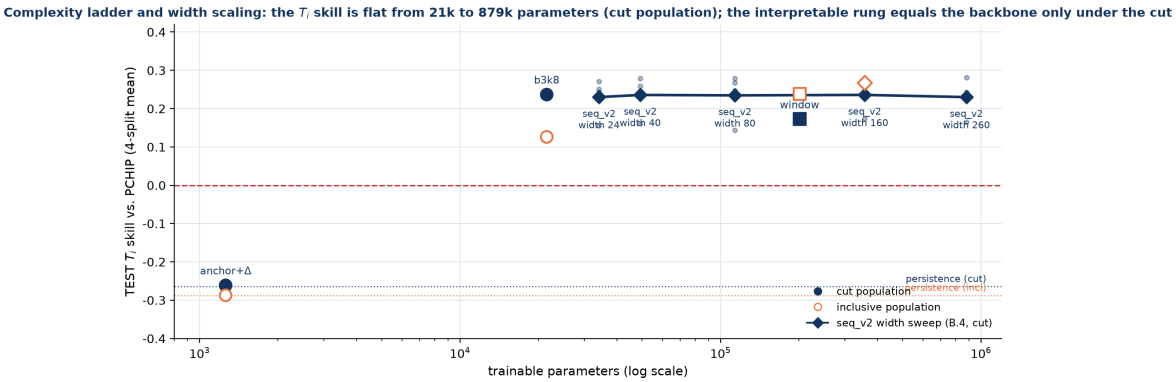


Figure 6: Complexity ladder and width sweep on one axis (TEST T_i skill vs. trainable parameters, 4-split means). Under the cut the 21k latent-bottleneck model sits on the backbone's curve and the curve is flat from 34k to 879k parameters; in the inclusive population the bounded-correction rung drops to +0.126.

(no targets touched) show the T_i latent encodes the last observed T_i (R^2 0.47–0.75) and the ECEI T_e proxy (0.31–0.48), distributed across dimensions, and barely encodes local BES activity (0.09–0.13) or staleness (0.01–0.07); the learned correction accounts for 25–39% of prediction variance. What the task needs is the full-grid causal state, and that state is small.

- **Without the cut the rung costs -0.16 to -0.21 against the backbone (4/4 significant) and sits below the windowed control on 3/4 splits**—not because eight latents are too few, but because a bounded correction cannot recover a spiked carried value: rows whose persistence error exceeds 2 keV are 0.6–1.3% of the inclusive test rows and carry 73–83% of `b3k8`'s T_i squared error (and 70–83% of every other arm's, PCHIP included). The rung condition still holds there (4/4 vs. the anchor); the backbone-tolerance condition is cut-conditional, and we state it so.

- **More capacity buys nothing.** Widening the `seq_v2` T_i encoder from 24 to 260 units (34k / 49k / 114k / 358k / 879k parameters; every other setting fixed; TEST once per point) gives mean T_i skill +0.230 / +0.236 / +0.235 / +0.236 / +0.230 (cut), paired against the 358k backbone within ±0.008 on average, the widest point significantly better on 1/4 splits, and no width down to 24 units significantly worse on $\geq$3/4; V_{rot} (branch fixed) does not move (+0.250 to +0.254). Together with the rung, this closes the model-size axis: the T_i information in {100 Hz BES/ECEI/MC + CES history + time} is exhausted by ∼50k parameters of causal recurrent state, and the remaining variance across splits (+0.14 to +0.17 on seed 42 vs. +0.23 to +0.28 on seed 123) is split variance, not capacity.

6.10 The edge concentrates in high-variability neighbourhoods

Using a target-independent, input-side proxy for local activity (large neighbour-bracket slope and/or high local CES-neighbour variance, computed *excluding* the target row), we split each TEST split into "peak" (high-local-activity; 4.1–4.6k T_i rows and 2.4–2.9k V_{rot} rows per split) and bulk rows and score `seq_v2` in each with the shot-clustered bootstrap (Fig. 7):

- T_i: peak-stratum $\text{skill}_{\text{PCHIP}}$ = +0.45 to +0.61 (cut) and +0.62 to +0.72 (inclusive), **PASS on 8/8**, against an interpolator that has the future neighbour; the bulk carries +0.09 to +0.20 (cut, PASS 4/4) and +0.06 to +0.19 (inclusive, PASS 2/4). Interpolation is near-optimal on the smooth bulk; the model's value is in the active stretches, and that statement is unconditional.

- V_{rot}: peak-stratum skill +0.54 to +0.79 (point estimates positive on 8/8, PASS 2/4 in each population, against persistence +0.75 to +0.86 PASS 8/8), and ≈ 0 in the bulk (-0.07 to +0.15, PASS 0/8). The asymmetry is therefore *regional*: globally V_{rot} ties interpolation, but in high-activity neighbourhoods—where smooth past+future interpolation is worst—even the history-driven V_{rot} predictor adds value. We report it as a regional strength with a thin significance base, not a reversal of the global null.

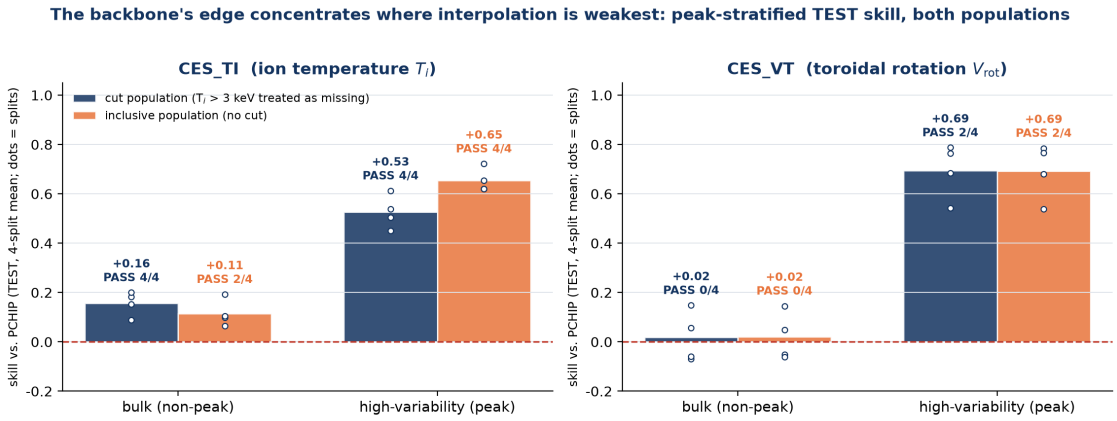


Figure 7: Peak-stratified TEST skill of `seq_v2`, both populations (4-split means, per-split points, PR4 counts). The advantage over interpolation concentrates where interpolation is weakest; V_{rot} skill lives entirely in the peak stratum.

Figure 8 shows the same statement on two held-out discharges of split 42, chosen by a rule fixed before plotting: the case-study discharge #31815 and the TEST discharge with the smallest within-discharge T_i spread. That spread is the per-discharge covariate that predicts where the model wins (Spearman $\rho = +0.40$ with the per-discharge skill against the causal GP, pooled over the splits). Where T_i moves (spread 220 eV) the nowcast beats PCHIP by +0.42 in MSE and ties the causal GP (+0.03). Where it does not (spread 34 eV, #31089) the nowcast follows the photon noise of the last sample and loses to both (−1.06, −0.73), which is the median-row cost §6.11 measures on every split.

6.11 The aggregate skill is a tail effect: in the median row the nowcaster carries its anchor

An MSE-based skill is decided by a sum of squared errors, and that sum is decided by a few large ones: the backbone's top 1% of T_i squared-error rows carry 48% (cut) and 83% (inclusive) of the total (PCHIP: 51% and 87%). "+0.24 on T_i" is therefore not, by itself, a statement that the model beats interpolation on a *typical* row. To separate the two we re-scored the frozen TEST artifacts, with no retraining, under three robust co-metrics whose rules—and the sentence each outcome would write—were fixed before any number: a median-absolute-error skill $M_1 = 1 - \mathrm{medAE}_{\mathrm{model}}/\mathrm{medAE}_{\mathrm{PCHIP}}$ (shot-clustered weighted-median bootstrap), the skill of a Huber loss clipped at the target's own 10 ms reproducibility bound (T_i 129.9 eV, V_{rot} 16.4 km/s), M_2, and a symmetric 1%-trimmed MSE skill M_3.

The T_i advantage comes from the tail. M_1 against PCHIP is negative, and significantly so, on all 16 population × target × split cells (cut T_i −0.198, inclusive −0.562). On half of the rows the model ties or trails PCHIP; the quantile ladder of the error ratio model/PCHIP falls from 1.20 at q50 through 1.01 at q70 and 0.96 at q75 to 0.78 at q99—the win begins in the upper error quartile, the fit-failure, transient and ELM rows. In that tail M_2 (+0.156) and M_3 (+0.210) are both significantly positive, so it is not an effect of the top 1% alone. The median row has a simpler picture: the backbone's 47 eV is persistence's 46 eV and the target's own bulk scatter of 46.4 eV. *On a typical row the nowcaster carries the last observation, noise included*, and PCHIP is 20% better there because averaging with the future neighbour halves that noise—a gain no causal method can have. The bulk ordering (acausal GP 38.5 < PCHIP 39.4 < causal GP 44.4 < persistence 46.1 ≈ backbone 47.2) is exactly the ordering of how many noisy samples each method averages.

The V_{rot} null is not a tail artifact. The pre-registration wrote "ties in aggregate but wins in the median" down as V_{rot}'s possible rescue. The opposite was observed: $M_1 = -0.488$ (cut) and −0.518 (inclusive), significantly negative on 4/4 splits, and in the median row the model is worse than persistence (4.3 vs

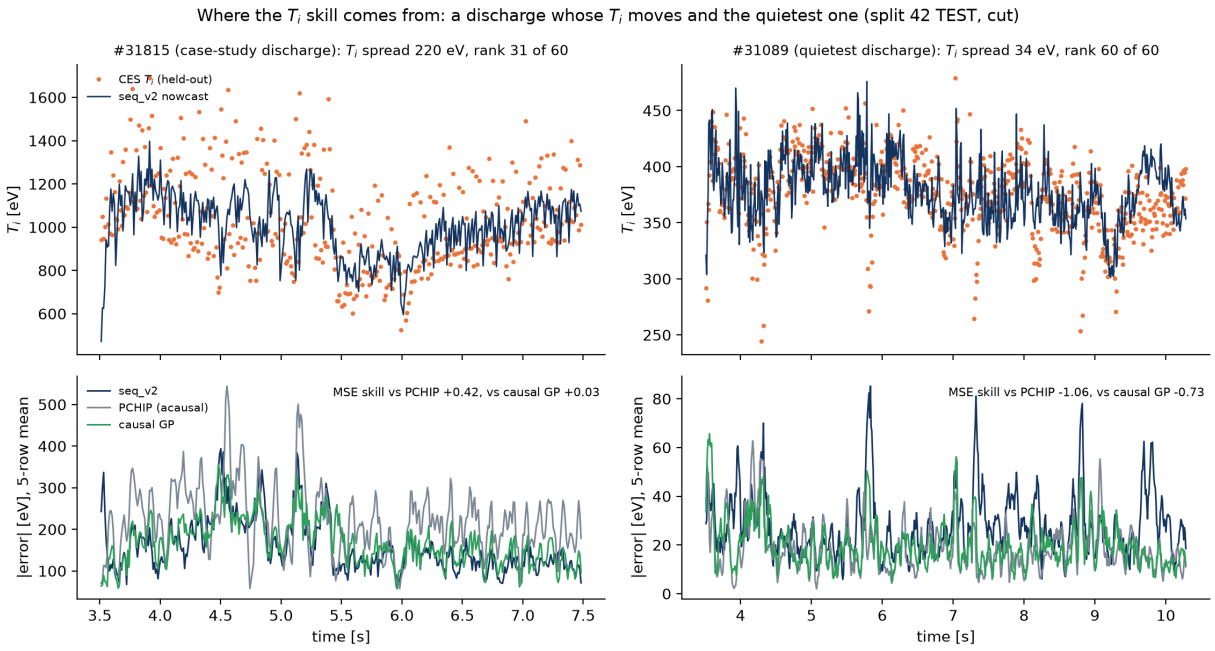


Figure 8: Two held-out discharges (split 42, cut population). Top: CES T_i and the `seq_v2` nowcast. Bottom: absolute error of `seq_v2`, PCHIP and the causal GP on the same rows (5-row running mean). Left: the case-study discharge #31815 (T_i spread 220 eV, rank 31 of 60). Right: the quietest TEST discharge (34 eV).

3.7 km/s). The aggregate tie is a bulk loss offset by a tail win (peak-stratum +0.54 to +0.79, §6.10), the same conclusion as the term-by-term audit of §6.7.

The loss's influence function moves that bulk. A loss that 1% of the rows decide teaches nothing about the other 99%. So *one* variable was changed—the influence function of the training loss—with the residual scale δ pinned to the target's own 10 ms reproducibility bound (cut: T_i 129.9 eV, $V_{\rm rot}$ 16.4 km/s; inclusive 274.3 eV): a Huber loss (r^2 inside, $2\delta|r| - \delta^2$ outside) and a Student-t negative log-likelihood at the same scale ($\nu = 3$, redescending). Both are normalized to coincide with r^2 inside δ, so they share the MSE control's bulk curvature and penalty ratio, and validation early stopping was kept on MSE for all three arms. Everything else is the backbone's; the control was not retrained. In the cut population Huber lowers the T_i median from 47.2 to 43.4 eV (M_1 vs the backbone +0.081 [+0.068, +0.094], significant on 4/4 splits) and $V_{\rm rot}$ from 4.3 to 3.8 km/s (+0.104, 3/4) while the aggregate MSE skill ties (paired −0.004 / −0.007, significant deficits 0/4). It is the first arm whose median beats persistence (+0.058, 3/4) and ties the causal GP, and it halves the median gap to PCHIP from −0.198 to −0.101. The Student-t arm is behind Huber on every metric and loses T_i MSE significantly on one split: a redescending influence discards useful tail signal. In the inclusive population the same arms came out behind on aggregate MSE (significant deficits 2/4), and the cause is the selection rule, not the loss: 7 of the 8 runs stopped with their best val-MSE checkpoint at epoch 2–4. A loss that ignores residuals beyond 274 eV has no reason to lower a validation MSE those residuals dominate. Selecting each robust arm on its *own* validation objective instead confirmed this: the inclusive arms then train 30–60 epochs, the Huber T_i median gain grows to $M_1 = +0.285$ [+0.250, +0.322] (4/4) and $V_{\rm rot}$ turns to +0.145 (4/4), with the $V_{\rm rot}$ median at 1.02× persistence—but one split keeps a significant T_i MSE deficit (−0.040; pooled −0.019), so by the pre-registered letter the rung stays conditional. By the pre-registered rule the Huber arm is recorded as the **robust rung, conditional on the cut population**, and the backbone keeps the MSE loss. The deployment statement now carries two numbers—aggregate skill (the tail) and median error (the bulk)—and the Huber arm keeps the first while buying the second.

Table 11: The error of a typical row. Four test splits pooled (301 shots), cut population, absolute-error quantiles in physical units. M_1 is the median-AE skill against PCHIP (shot-clustered 95% CI); the win rate is the fraction of rows on which the model's error is below PCHIP's. The backbone's T_i median of 47 eV equals the target's own 10 ms bulk scatter (MAD of adjacent-sample differences, 46.4 eV). The Huber arm is the subject of the last paragraph of §6.11.

target	arm	median	q75	q90	M_1 vs PCHIP	win rate
T_i [eV]	`seq_v2` backbone (MSE loss)	47.2	101.0	202.4	−0.198 [−0.259, −0.135]	0.48
	`seq_v2` Huber loss	**43.4**	**93.8**	**193.3**	−0.101	
	persistence	46.1	116.6	262.7		
	causal GP	44.4	106.7	219.8		
	PCHIP (uses the future)	39.4	105.4	236.6		
$V_{\rm rot}$ [km/s]	`seq_v2` backbone (MSE loss)	4.3	10.0	21.4	−0.488 [−0.604, −0.387]	0.40
	`seq_v2` Huber loss	**3.8**	**9.2**	**19.8**	−0.333	
	persistence	3.7	9.3	20.9		
	causal GP	3.8	9.3	20.0		
	PCHIP (uses the future)	2.9	7.8	18.3		

6.12 A physics filter is the median estimator: a dual readout

§6.11 located the headroom in the median row, where the backbone only equals persistence. That row is the regime a filter is built for: a slowly varying level observed through noise, with occasional jumps. We therefore asked whether a *physical* measurement model, not a learned one, owns the median. The cell is a scalar state-space filter per target, run causally on the CES history alone: the prior holds the level and lets it jump at rate λ with jump variance σ_J^2, and the measurement is Student-t ($\nu_0 = 3$) at the target's 10 ms bulk scatter estimated on TRAIN (T_i 0.116 / 0.071 target s.d. in the cut / inclusive population, $V_{\rm rot}$ 0.060). Every parameter is fixed by the registration; training it end-to-end moves none of them by more than 7%, so they are not reported as measured plasma quantities.

How the measurement model was found. Each step was one pre-registered variable. An Ornstein–Uhlenbeck prior with Gaussian noise moved the T_i median below persistence for the first time (0.98× cut) but left $V_{\rm rot}$ at 1.19×; a hold-or-jump prior fixed T_i (0.98× / 1.00×) and left $V_{\rm rot}$ at 1.18× under both priors. A TRAIN-only diagnostic located the cause: $V_{\rm rot}$ increments do not trend, and the Gaussian noise scale had been set at the tail, six times the bulk scatter, so the filter shrank every fresh sample. Student-t noise at the bulk scale removes it: the pure filter sits at **0.916**× / **0.911**× persistence for T_i and **0.975**× / **0.976**× for $V_{\rm rot}$.

One output cannot carry both objectives. Adding the backbone's residual ($y = a + r$, MSE-trained) gives back aggregate skill and meets the physics-rung bar in the cut population (M_1 vs the backbone +0.077 to +0.117, 4/4, MSE deficit $\leq 1/4$), but misses the inclusive MSE guard by 0.0014—and it gives the anchor's median back ($V_{\rm rot}$ 1.08–1.12×, inclusive T_i 1.06×). Two further single-variable rows tried to keep it: selecting the residual's checkpoint on validation median error ($V_{\rm rot}$ 1.07×; the passing population swaps, inclusive in and cut out), and weighting the residual by the filter's variance gain $k = s^2/(s^2 + \sigma_r^2)$ ($V_{\rm rot}$ 1.15–1.21×: the residual learns to undo k, imprecisely). Within this family it is therefore measured, not assumed, that an output which must also win the aggregate moves the quiet rows.

The dual readout. The pre-registered consequence is to report the median and the mean from separate estimators on the same rows (Table 12). For T_i the physics filter is the median estimator **unconditionally**: it beats the backbone and persistence on 4/4 splits in both populations, at 0.906–0.920× persistence with a split spread of ±0.007, against ±0.1 or more for every trained arm. For $V_{\rm rot}$ it beats the backbone in the median on 4/4 splits in both populations and ties persistence (1/4 significant, 0.94–1.00× by split). The tie survives fitting the filter on TRAIN. Likelihood-fitted jump statistics (5–13 Hz) and a likelihood-fitted noise law (Student-t with $\nu \to 2$ and a 2.2 km s^{-1} core) leave the median at 0.97–0.98× persistence, with persistence

Table 12: Dual readout on the frozen TEST rows (no retraining). Median estimator: the pure Student-t hold-or-jump filter; mean estimator: the trained model (anchor + residual where its MSE guard holds, i.e. cut; `seq_v2` in the inclusive population). M_1 is the median-AE skill, shot-clustered paired bootstrap; counts are splits with $M_1 \geq +0.05$ and CI > 0 (vs the backbone) or $M_1 > 0$ and CI > 0 (vs persistence). Ranges are over the four splits.

target	population	median / persistence	M_1 vs `seq_v2`	M_1 vs persistence	MSE vs `seq_v2`
T_i	cut	0.911–0.920	+0.06 to +0.14, **4/4**	+0.08 to +0.09, **4/4**	−0.30 to −0.41
	inclusive	0.906–0.915	+0.22 to +0.39, **4/4**	+0.09, **4/4**	−0.35 to −0.47
$V_{\rm rot}$	cut	0.939–1.000	+0.09 to +0.23, **4/4**	0.00 to +0.06, 1/4	−0.06 to −0.51
	inclusive	0.938–1.003	+0.09 to +0.27, **4/4**	0.00 to +0.06, 1/4	−0.06 to −0.52

beaten on at most 2/4 splits. The reason is measured. The 10 ms scatter sits at the photon floor: the ratio of the relative T_i scatter to the Doppler velocity scatter is the $\sqrt{2}$ of a Gaussian line fit in 8/8 split–population cells, and the noise of the two channels is independent. Between labels the filter's predictive is symmetric about the carried value, so its median-optimal point is the last value. The tie is therefore a limit of the 10 ms CES measurement, not of the estimator. The filter loses the aggregate MSE to the backbone (−0.06 to −0.52) because event rows stay late, which is exactly why it is not the mean estimator. The deployment statement becomes: median from the filter, mean from the nowcaster, and the trade between them measured, not hidden in one output.

6.13 The predictive distribution: one heavy-tailed peak, reported calibrated

A nowcast used for control needs an uncertainty, so the backbone's head was replaced by a predictive distribution trained by its negative log-likelihood (nothing else changes; the posterior mean goes through the same point harness). Three likelihood families were registered in turn: a heteroscedastic Gaussian, a two-component Gaussian mixture, and a Student-t with learnable ν. The reference is the backbone's point prediction with its own TEST RMSE as the scale, an oracle choice that favours the reference. The registration fixed four guards (CRPS gain, central 50 % and 90 % coverage, and no posterior-mean MSE loss against the backbone) and named the arm before any number.

The Gaussian is rejected. Its central 50 % interval covers 0.62–0.80 of the rows (Fig. 9): a single Gaussian scale inflated by the tail rows is too wide for the bulk. On $V_{\rm rot}$ its likelihood weighting is pulled by the $|v| > 1000\,{\rm km\,s^{-1}}$ fit failures and its mean loses MSE to the backbone on 3/4 cut splits.

The named arms are calibrated. In the cut population the Student-t head ties the mixture (CRPS difference +0.009, below the 0.02 tie width) and is named for having fewer parameters. It beats the reference in CRPS by +0.229 (T_i) and +0.363 ($V_{\rm rot}$), significant on every split, with 50 % / 90 % coverage 0.48–0.51 / 0.91–0.92. In the inclusive population only the mixture passes the guards (+0.444 / +0.351, coverage 0.53–0.55 / 0.90–0.91). The t location there is a redescending estimator: on the > 3 keV fit-failure rows it stays at the bulk and pays their full square in MSE (−0.134 on the peak rows), whereas the mixture moves its mean with the wide component.

What the distribution says about the physics. The mixture never uses two peaks: its two-mode test fires on no row, and the second component is a wide one with weight ≈ 0.07. The fitted t tail is $\nu = 2.8$–3.5 for T_i and sits on the $\nu > 2$ floor for $V_{\rm rot}$. The nowcast error is therefore one heavy-tailed peak: a narrow photon-noise core with rare large departures from events and fit failures, the same shape the physics filter needed for its measurement model (§6.12). The T_i predictive width rises 1.2× on event rows; the $V_{\rm rot}$ width does not grow with the time since the last label, so the $V_{\rm rot}$ uncertainty is a per-discharge scale, not a variance that accumulates between labels. Figure 10 shows the intervals on one held-out discharge.

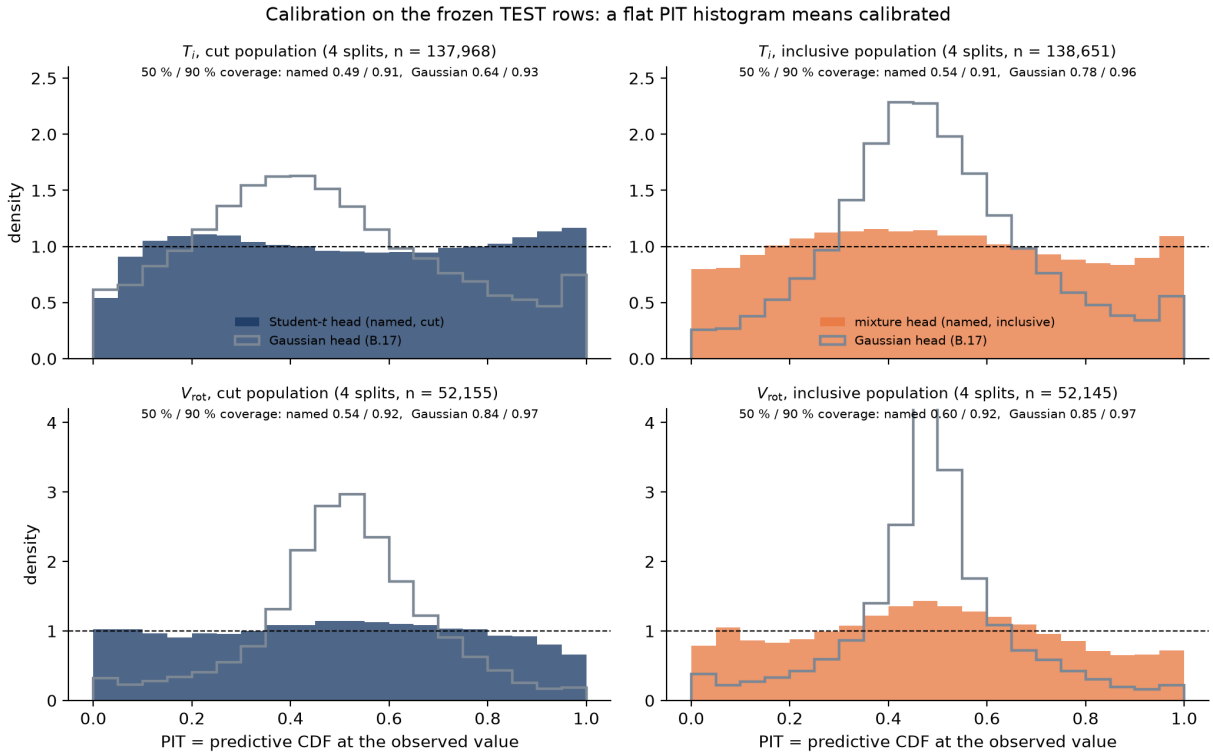


Figure 9: PIT histograms on the frozen TEST rows, four splits pooled. Filled: the registered arm (Student-t in the cut population, mixture in the inclusive one). Outline: the Gaussian head. A flat histogram at density 1 is calibrated; a central hump means intervals that are too wide in the bulk.

6.14 Model families and ensembles: the inductive-bias axis closes too

§6.9 closed the size axis within one family. Two model-side questions remained: whether a different *inductive bias* extracts more from the same inputs, and how much an initialization ensemble of the same model buys. Both were measured with the rule written first.

The 4-init ensemble lands exactly on the tie threshold. Averaging the four initializations' normalized predictions per split on the confirmed-protocol B.1 grid (4 splits × 4 inits) gives paired T_i gains of +0.011 / +0.010 / +0.027 / +0.030 in the cut population (4/4 significant, mean +0.020) and a mean of +0.007 (2/4) in the inclusive one. The pre-registered rule made a deployment recommendation only above the 0.02 tie band, so this is recorded as a **tie** and the headline stays a single run. The +0.048 once measured in a reduced configuration was that configuration's initialization noise.

An offline model-family card. The inductive-bias comparison was designed as a *card*, not a hypothesis test: one hypothesis per family, scored against an offline control that differs from `seq_v2` only in direction (its bidirectional version, a BiLSTM) on identical masks, splits and budget. The setting is offline gap-filling: observed CES rows are hidden in spans drawn from the measured missing-run-length distribution and filled (three quarters of the scored points sit in gaps longer than 10 steps). Three arms: a neural process with a shot-level latent and cross-attention (`np`; hypothesis—the $V_{\rm rot}$ gain concentrates where the same shot has few labels), and the transformer backbone of the public tokamak foundation model TokaMind, pretrained on another device (MAST), transplanted (`tm_pre`) against the same architecture from a random initialization (`tm_scratch`; hypothesis—their difference is the value of cross-device domain knowledge). 4 arms × 4 splits × 2 populations; TEST scored once per arm.

Three things are read off Table 15. *First*, on $V_{\rm rot}$ the family ties: the two cut-population edge cells have

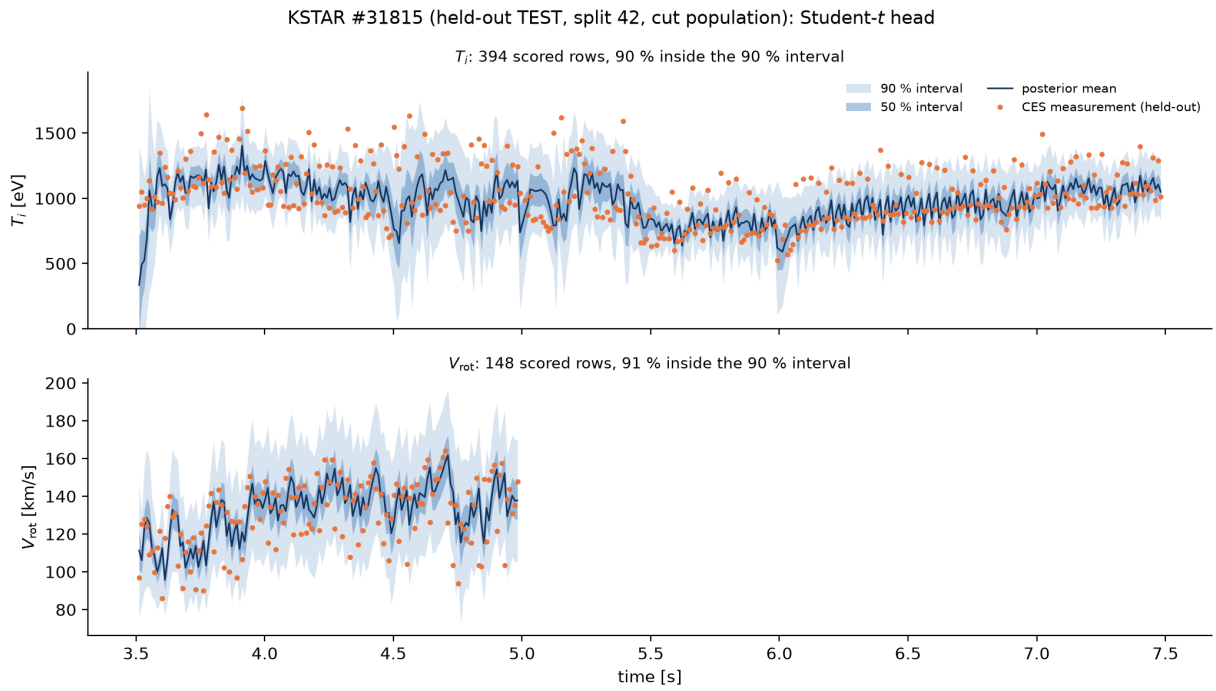


Figure 10: Posterior mean and central 50 % / 90 % predictive intervals of the Student-t head on the held-out discharge #31815 (split 42, cut population), with the CES measurements. The T_i axis is cut at 0; the first row's interval extends below it.

CI lows of 0.000 and 0.001, all three arms tie in the inclusive population, and no stratum reproduces either cell. *Second*, the neural process's hypothesis is rejected exactly where it was built to hold: with no same-shot V_{rot} label *no* arm beats the training mean and `np` is the worst, and with 3 and 10 labels all four arms rise by the same amount (+0.15 → +0.40). What carries a shot's operating regime is its context labels, not an encoder, and the 28% between-shot V_{rot} variance belongs to operating variables the diagnostics do not contain (beam sources, I_p, B_T)—the next measurement is `np` conditioned on that metadata, not another encoder. *Third*, the pretrained and random backbones agree to 0.004–0.013 on every cell: what was learned on MAST transfers nothing to KSTAR CES gap-filling, the same sign and size as TokaMind's own profile fine-tuning result (−0.005). The transformer family is also behind the BiLSTM on T_i at 7× the parameters (cut −0.09, inclusive −0.03 to −0.04). The `np` T_i +0.026 clears the rule in the cut population only and ties in the inclusive one, so it is recorded as a marginal effect of the same size as the ensemble's.

The card's one positive result is the gap-length profile. On V_{rot} all four learned arms lose to PCHIP in one-step gaps (−0.25) and win by +0.4 to +0.6 in gaps of two steps or more (the acausal GP: +0.02 to +0.08). The scale at which V_{rot} is recoverable from the diagnostics is 100 ms; the 10 ms scale, where the adjacent CES sample is a better estimator than anything the diagnostics say, is where the V_{rot} tie of §6.7 lives. The offline gap-filler and the causal nowcaster are the same model seen at two scales, and a V_{rot} claim has to say which.

6.15 Cut-threshold sensitivity

Retraining the backbone with the spike threshold at 2.5 and 4 keV (four splits each, scored in its own population) gives mean T_i skill +0.230 and +0.232 against +0.236 at 3 keV, PR4 4/4 vs. PCHIP and 4/4 vs. the causal GP at every threshold, and V_{rot} +0.252 / +0.253 / +0.257. The threshold is immaterial across the range a physicist would defend; what matters is the two populations, not where inside the tail the cut falls.

Table 13: Offline model-family card. Skill vs PCHIP (4 splits pooled, shot-clustered 95% CI) and the paired difference δ to the BiLSTM control (normalized by the PCHIP MSE). Verdicts follow the pre-registered rule: $\delta > 0.02$, CI low > 0, same sign on $3/4$ splits, and the same against the acausal GP. The label-sparsified sets keep k V_{rot} labels in label-rich test shots; at $k = 0$ the reference is the training mean.

population / target	metric	BiLSTM (control)	`np`	`tm_pre`	`tm_scratch`
cut T_i	vs PCHIP	+0.274	+0.300	+0.179	+0.184
	δ vs control		+0.026 [+0.004, +0.049] improve	−0.094 worse	−0.090 worse
cut V_{rot}	vs PCHIP	+0.364	+0.384	+0.376	+0.384
	δ vs control		+0.021 [+0.000, +0.050] edge	+0.012 tie	+0.021 [+0.001, +0.044] edge
incl. T_i	vs PCHIP	+0.410	+0.403	+0.371	+0.378
	δ vs control		−0.006 tie	−0.039 worse	−0.032 worse
incl. V_{rot}	vs PCHIP	+0.336	+0.327	+0.321	+0.333
	δ vs control		−0.009 tie	−0.016 tie	−0.003 tie
V_{rot}, $k = 0$ labels	vs train mean	−0.025	−0.040	−0.016	−0.003
V_{rot}, $k = 3$ labels	vs PCHIP	+0.146	+0.148	+0.132	+0.137
V_{rot}, $k = 10$ labels	vs PCHIP	+0.407	+0.403	+0.392	+0.398

Table 14: Recurrent family, pooled rescoring over 301 discharges. "Win rate" is the fraction of discharges on which the model beats the causal GP; "−top10" is the pooled skill after deleting the ten largest contributors. The trend is +0.0497 per decade of context [+0.0363, +0.0638]—rising.

Context	T_i skill vs causal GP [95% CI]	Win rate	−top10
20 ms	+0.057 [+0.027, +0.085]	**0.52**	+0.028
30 ms	+0.087 [+0.061, +0.111]	0.60	+0.060
50 ms	+0.104 [+0.079, +0.128]	0.64	+0.077
70 ms	+0.119 [+0.095, +0.142]	0.66	+0.092
150 ms	+0.123 [+0.096, +0.148]	0.66	+0.096
630 ms	+0.143 [+0.118, +0.168]	0.67	+0.116

6.16 Context, structure and cost: what sets skill and what sets the price

Everything above scores one architecture at one context length. This section moves the two separately and asks which one produces the result. They answer very differently: stretching *context* from 20 ms to 630 ms raises T_i skill from +0.055 to +0.138, while swapping the *operator family* at matched context—recurrence for dilated convolution or for attention—never moves it by more than 0.023 anywhere. The architecture should therefore be chosen on cost rather than on skill, and that cost follows the number of dispatched operators rather than the parameter count.

Reach must be measured by training, not by truncation. Truncating a trained backbone's recurrent state to r steps after the fact makes T_i look as though it loses out to 500 ms, but that reads *unlearned state reconstruction* as missing information. Retraining on non-overlapping segments of length r recovers at least 84% of that deficit. Every reach number below is from a model trained at that context, not from a truncated one.

Saturation is near 50 ms, and what context buys is typicality rather than the average. Applying the pre-registered rule—the smallest r whose deficit against full context is under 0.02 with at most one significant deficit out of four splits—to a ladder densified to one-step spacing returns **5 steps = 50 ms**. That threshold must not be read as a crossing point. Pooling the four splits by discharge and rescoring over 301 discharges gives Table 14, and **the model beats the causal GP at 20 ms as well** (+0.057, 95% CI [+0.027, +0.085]).

The win-rate column is this section's conclusion: 0.52 → 0.60 → 0.64 → 0.66, then flat from about 70 ms. At 20 ms the model wins on average but on barely half the discharges; by 70 ms it wins on two thirds. **Context does not raise the average so much as it makes the win typical.** That is the statement a deployment audience needs, and the threshold must not be described as "the context needed to be useful": already at 20 ms the model beats persistence—what a deployed system actually displaces—by +0.356.

Table 15: Per-family reach ladders (T_i skill vs the causal GP, mean of four splits · number of splits whose CI clears zero). On this work's own promotion bar of $\geq 3/4$ all three families are monotone. The convolutional arm's 30 ms rung is its structural minimum (one layer, receptive field 3), so its crossing is an upper bound.

Context	Recurrent (LSTM)	Dilated conv (TCN)	Banded attention
20 ms	$+0.055 \cdot 2/4$	—	—
30 ms	$+0.083 \cdot 3/4$	$+0.079 \cdot 3/4$	—
50 ms	$+0.098 \cdot 3/4$	$+0.103 \cdot 3/4$	$+0.090 \cdot 2/4$
70 ms	$+0.115 \cdot 4/4$	$+0.111 \cdot 4/4$	$+0.094 \cdot 3/4$
150 ms	$+0.117 \cdot 4/4$	$+0.130 \cdot 4/4$	$+0.119 \cdot 4/4$
630 ms	$+0.138 \cdot 4/4$	$+0.124 \cdot 4/4$	$+0.122 \cdot 4/4$

Table 16: Online single-step cost by family. R is reach and "ops" counts dispatched `aten::` operators per step. Attention is also $O(1)$ in R thanks to the KV cache; it simply carries a 3.5–4.3× worse constant.

Family	Cost in R	$R = 15$	$R = 63$
Recurrent (LSTM)	$O(1)$ — the state carries it	111	111
Dilated causal conv	$O(\log R)$, +48 per layer	209	305
Banded attention	$O(1)$, constant 4.3×	473	473

One methodological caution follows from this ladder. The per-split 4/4 count is not a monotone function of reach at one-step spacing ($2/4 \to 3/4 \to 4/4 \to 3/4 \to 4/4 \to 4/4 \to 3/4 \to 4/4 \to 4/4 \to 4/4$ from 20 to 630 ms) while the point estimates underneath rise smoothly. That is what a five-level vote on four samples does, not a problem with the runs. Report the counts; decide on the deficit.

At matched context the operator family does not set skill. Holding reach fixed and comparing a recurrent LSTM (357,570 parameters), a dilated causal convolution (3,238) and banded attention (295,746), all three beat the causal GP 4/4 at +0.113, +0.119 and +0.119. A 110× parameter range and three of the most dissimilar sequence operators available produce no distinguishable difference. Read as ladders this is Table 15; paired at matched reach—the more robust contrast—the differences are quieter still: `tcn3/5/7` tie the same-reach LSTM (−0.004, +0.005, −0.004, with no significant split either way) and only attention at the 70 ms rung returns a difference under the pre-registered rule (−0.023, significant on 3/4). The largest family gap anywhere is 0.023 in T_i, against +0.060 for moving reach from 20 to 70 ms.

A fourth family, a diagonal state-space model, behaves differently on the same ladder. It does convert context into skill (+0.0235 per decade [+0.0102, +0.0372]) but stops around 70 ms at a lower ceiling: +0.105 at 630 ms against the recurrent +0.143, with a win rate capped at 0.58–0.60 rather than 0.66. It is, however, the best pooled point of any family at 20 ms (+0.065 against the recurrent +0.057)—a same-rung pooled comparison rather than a paired test, so it is reported and not promoted. V_{rot} declines with context in **all four** families.

So the architecture is chosen on cost, and cost is operator count. Online single-step latency tracks the number of dispatched operators (about 2–3 μs each) rather than parameter count or arithmetic. Those counts are exact on any machine; this machine's absolute timings moved by 21.84× between sessions and are therefore not quoted.

On this problem attention is *strictly dominated*: `tcn3k` (3,238 parameters) matches its skill (+0.119, 4/4) with 91× fewer parameters and 2.3–2.7× fewer dispatched operators. Three operating points are worth naming: `tcn3k` matches the 357,570-parameter backbone on T_i and beats the causal GP 4/4, `v2m4k` (3,898) is the cheapest recurrent arm that still clears 4/4, and `tcn2k` (1,808) is the cheapest arm of any family. They are not "the small version of the model"; they are the answer to what this problem actually requires. The 10 ms budget binds no arm, and the 1 ms verdict is **suspended** by the pre-registered rule because of the between-session spread above.

Why the windowed family could not reach this axis. §6.8 reports that one past *observation* is enough and this section reports that about 50 ms of *contiguous context* is needed. They do not conflict, because they count different resources. The windowed family satisfied the first and could not reach the second at any W, by construction: average-pooling the window discards its ordering, the temporal-subset augmentation makes each window an arbitrary non-contiguous subset of past rows, and rows without an observed target were dropped before the model saw them. The numbers land exactly where that explanation predicts: the W=2 window model scores +0.041 (1/4) against the causal GP and `seq_v2` truncated to the same two steps scores +0.055 (2/4). Different architectures, same starvation, same result. The full-grid reframing was not a modelling preference; it was the only way to make the resource that matters reachable.

The model wins on discharges where the target moves. Pooled rescoring produces a V_{rot} result that would be badly misread on its own: every rung's pooled CI clears zero (at 70 ms, +0.123 [+0.036, +0.248]). But the per-discharge win rate is 0.48, the median discharge's skill is about zero, and removing the top five of some 62 discharges takes the pooled advantage to zero or below on all four splits. **A tight interval is not the same as a general effect.** T_i is the opposite: it wins on 70% of discharges and its single largest contributor accounts for only 9–16%.

Of eleven covariates tested for predicting the win, one survives, and it is *how much the target moves within a discharge*. Split into terciles of that spread, T_i wins 42% / 83% / 85% and V_{rot} wins 34% / 48% / 55%. The mechanism is physical rather than statistical: where a discharge is flat the causal GP is already near-optimal, since interpolating a straight line is easy, and the model has nothing to add. That V_{rot} reaches only 55% even in the most variable tercile restates the term-by-term audit of §4.2 from the other side—the driver of rotation is unobserved, so the model cannot track motion it has no input for. The family preferences point the same way: V_{rot}'s best arm is the small *recurrent* one and T_i's is the small *convolutional* one, exactly as the routing predicts, because V_{rot} rides a highly autocorrelated carried value where recurrence suffices while T_i integrates fast-diagnostic history, where weight sharing across time pays.

7 Model-selection protocol

Every model decision in this paper was taken on validation data or under a decision rule committed in writing before the corresponding TEST scoring, and TEST was scored once per decision.

The windowed family. Its architecture was reached by a sequence of controlled experiments, each *one* change that had to preserve the data contract, scored on the *clean, non-augmented* validation $\text{skill}_{\text{PCHIP}}$—never the augmented validation loss, which rewards smoothing exactly where interpolation is already strong—and kept iff it improved the best score so far. Its history length was then set by the sweep of §6.8, and holds were removed from training by the audit of §3.4.

The backbone gate. The sequence framing was adopted only after the four-condition gate of §6.3 (sign on 4/4 splits, pooled run-cluster CI excluding zero, sign preserved under budget equalization, no V_{rot} deficit) had been fixed and then met. Under the same rule the one architectural candidate explored afterwards—observation-masked causal attention over each target's own past observation steps, added to `seq_v2` with zero-initialized projections—was positive on 4/4 splits (+0.009, +0.013, +0.033, +0.020 vs. the backbone) but significant on 1/4 against a pre-committed bar of $\geq 3/4$, so it was *not* promoted; the backbone stayed `seq_v2`. Its validation gains had been 2/2 significant on the exploration splits, which is the usual optimism of selection-split results and the reason the promotion bar is set on TEST.

The rung and the width sweep (§6.9) had their two-branch verdicts and their descriptive readings (ceiling / knee) written down before any TEST scoring, and no backbone re-selection was allowed on the sweep by construction. The full pre-registration, its per-batch execution records and the runner scripts are in the released repository.

8 Is it deployable?

Two things stand between a skill score and a usable instrument: it has to run inside the measurement cadence, and it has to say how much to trust it.

Latency: stateful one-step inference fits on a CPU with room to spare. CES sits on a 10 ms grid, so an online gap-filler has 10 ms per step minus whatever acquisition and control already spend. A recurrent nowcaster is run online *statefully*: the hidden state is carried across the grid and each new row costs one recurrent step at batch 1, which is the number that matters. Measured as the network forward pass only (feature assembly is outside the model) over 1,000 timed calls after warm-up on an idle laptop-class machine, the `seq_v2` step costs **0.97 ms median and 1.49 ms at p99 on CPU** (15% of one grid period; p95 1.17 ms), and 1.08 / 1.79 ms on the GPU—at this size a GPU buys nothing at batch 1. The whole-segment re-run that offline scoring uses costs 2.6 ms median / 4.1 ms p99 for a 100-row (1 s) block and 6.1 / 9.6 ms for 300 rows on CPU, i.e. 38–49k rows/s of bulk reprocessing (46–62k rows/s on the GPU). The windowed control at batch 1 is slower: W=2 2.9 ms median and 4.5 ms at p99 on CPU (5.1 / 8.6 ms on the GPU); W=4 3.1 / 5.1 ms. Earlier idle-machine runs of the same benchmark put the `seq_v2` step at 0.51–1.05 ms median and 0.99–1.61 ms p99, and one showed a heavy CPU tail for the windowed control (18.9 ms p99), because a laptop's power states move these numbers; the ordering—sequence step $\ll$ windowed forward pass, CPU sufficient—did not change in any run. The practical guidance is therefore: *run the stateful nowcaster on the control computer's CPU*; it leaves more than 80% of the 10 ms budget to acquisition and control.

Uncertainty: distribution-free intervals without touching the model. The community standard for fusion profile data is Gaussian-process regression, which returns a posterior; our model returns a point. Fitting a variance or quantile head would require retraining and would move every number above—so we use *split conformal prediction* instead: calibrate on the run's own validation split, change nothing about the predictor, obtain distribution-free finite-sample marginal coverage. Two variants: a single quantile (*global*) and one per Δt bin (*Mondrian*). The identical procedure is applied to PCHIP and persistence, so what is compared is interval quality, not calibration technique. At $\alpha = 0.10$, TEST, both populations:

- **The model's intervals beat both baselines' in every cell** (32/32 population × target × variant × split) by the Winkler interval score, which penalises width and misses jointly: T_i 1,272 vs. 1,554 (PCHIP) vs. 1,727 (persistence) in the cut population, 2,290 vs. 2,851 vs. 3,120 inclusive; V_{rot} 150 vs. 164 vs. 179 (global variant, 4-split means). This does not follow from better point estimates; it is a separate property.
- **In the inclusive population the model's T_i intervals are actually *wider* than PCHIP's** (half-width 224–255 vs. 211–241 eV) and still score better: the spikes inflate the miss penalty, and the model misses less. Mondrian calibration tightens every T_i arm by ≈4–5%, leaves V_{rot} unchanged, and changes no verdict.
- **Honest failure: coverage is marginal, not conditional.** T_i coverage is 0.87–0.92 against a 0.90 target (one split under-covers for every arm) and V_{rot} 0.91–0.94; per-shot coverage still ranges widely. Conformal guarantees coverage under exchangeability, and calibration and test are disjoint *discharges*; shot-level shift breaks it. For a deployed instrument this matters more than the pooled number, and fixing it needs shot-conditional calibration that the present shot count cannot support.

9 Where the remaining headroom is

The negative results in this paper are unusually specific, and specificity is what makes them useful: they do not say "this problem is hard", they say *which change would move it.* Capacity is ruled out (§6.9), longer windows are ruled out (§6.8), inductive bias, cross-device pretraining and ensembling are ruled out (§6.14), and the framing that buys reach has been adopted (§6.3). The one model-side axis that moved is the loss's influence function, and its gain is a bulk gain (§6.11). What remains beyond that is data, and each lever is identified by evidence already in this dataset.

1. CES fit-quality metadata would replace a proxy with a measurement. The two-population report exists because a value cut is a proxy for "the fit failed": it removes one-sample upward events that no method can predict, but it is one-sided (dips are untouched) and it catches only 19% of $\geq 2\times$ upward outliers (§3.5). With per-sample fit χ^2 or signal level, a quality cut would replace or accompany the value cut in every arm at once, and the two populations would collapse into one. The route to those values already exists on the same device: NN-CES analyses CES spectra without beam modulation and so yields a fit diagnostic per exposure [18], which also removes the gaps that come from modulation itself. This lever is a CES reanalysis product, not a code change on our side. $V_{\rm rot}$'s own spikes (119 rows above 1,000 km/s) would be handled by the same rule; until then any persistence-anchored $V_{\rm rot}$ comparison reports the share of squared error those rows carry, as §6.9 does.

2. The Mirnov information was destroyed by preprocessing, not absent from the plasma. It would be wrong to conclude from §6.7 that magnetics carry no rotation information. On the identical 10 ms grid, lag-1 autocorrelation within contiguous blocks is +0.568 for BES and +0.572 for ECEI but −0.009 for the Mirnov coils, with 82% of blocks below $|r| = 0.1$: the magnetic channel is white noise *on this grid*. That is the signature of $\mathrm{d}B/\mathrm{d}t$ oscillating at the kHz mode frequency and being decimated to 100 Hz without an anti-aliasing filter, so successive samples have random relative phase. The mode-rotation frequency—the one quantity in our diagnostic set that could proxy toroidal rotation—was therefore discarded upstream of the model, which is also why every derived MC feature we tried (integral, PCHIP integral, $|MC|$, rolling RMS) failed: information already lost cannot be recovered downstream. The fix is a preprocessing change, not a model change: per-window RMS, band power, mode number and mode-rotation frequency computed from the *raw* kHz Mirnov timeseries, routed into the $V_{\rm rot}$ branch under a pre-registered pilot-then-expand rule. This is the highest-value experiment we can name for $V_{\rm rot}$, and it is testable on archived data.

3. The actuator channel is simply absent. Toroidal rotation is set by injected torque, and no torque signal exists in this dataset. The data show the break precisely: between shots, the ECE-derived T_e surrogate correlates with T_i at $r = +0.353$ ($p = 3 \times 10^{-17}$) but with $V_{\rm rot}$ at $r = +0.024$ ($p = 0.58$), and within shots the T_i correlation keeps a consistent sign while the $V_{\rm rot}$ one is sign-random. So T_e does track the heating level, and heating level does not reach rotation—power is not torque, which depends on beam energy, tangency radius and injection geometry. Adding NBI torque (or the per-beam power and geometry from which it is computed) is a data-acquisition task, not a modelling one, and it is the only lever we can identify that addresses the *cause* of rotation rather than its autocorrelation. A positive control already exists in the literature: recurrent full-discharge simulators on DIII-D that receive the beam actuators as inputs do predict rotation's evolution [52], so the missing channel is learnable once measured.

4. The rest of the bulk belongs to the future neighbour. After the Huber arm's recovery, a 4 eV median gap to PCHIP remains (§6.11). Its size is what averaging two adjacent noisy observations buys, so the only way a causal model closes it is by denoising the history—weighting it by the per-exposure fit quality a CES re-analysis would provide—which is the same deliverable as lever 1. And the zero-label row of the model-family card (§6.14) states lever 3 most sharply: with no same-shot $V_{\rm rot}$ label, no learned arm rises above the training mean from BES, ECEI and Mirnov alone.

None of these is a statement that the present result is at a ceiling. They are the places where the next measurable gain is, ordered by how cheaply they can be tried, and all three are actionable on archived KSTAR data.

10 Limitations

The headline skill is a tail statistic. In the median row the backbone equals persistence and trails PCHIP by 20%, and $V_{\rm rot}$ is below persistence in the median (§6.11); the paper therefore reports median absolute error beside every aggregate skill and does not claim to beat interpolation on a typical row. The scale-pinned Huber loss recovers 8–10% of the bulk at no aggregate cost, reported as conditional on the cut population; selecting on the arm's own objective lifts the inclusive median gain to +0.285 but leaves one split with a significant MSE deficit (§6.11). The T_i median is owned by a physics filter unconditionally

(§6.12), reported beside, not inside, the aggregate model; for V_{rot} that filter ties persistence in the median at the photon floor of the 10 ms line fit. **Statistical power.** The unit of replication is the shot; with 96 (T_i) / 60–66 (V_{rot}) test shots per split, power bounds every significance call, and per-shot squared-error differences are heavy-tailed (a few discharges dominate the bootstrap spread; in the inclusive population ≈1% of rows carry 70–83% of every arm's T_i squared error). **MNAR optimism.** Skill is measured on observed points only; the reweighting of §6.5 bounds the optimism for the causal comparison in both populations and for the offline comparison with population-conditional significance, and it covers 54–68% of missing T_i rows and 4–6% of missing V_{rot} rows. **The offline claim is bounded above by a GP tie.** The offline GP beats PCHIP and the model only ties it (1/8 cells significant), so "beats future-using interpolation" means "beats the pre-registered interpolants"; it does not extend to every offline method. It does extend to the causal GP, the strongest deployable one. **The value cut is a proxy.** Two populations bracket the fit-failure question; a fit-quality cut would settle it (§9), and V_{rot}'s spikes remain uncut. **Campaign transfer rests on one temporal block** (§6.6): four initialisations, not four splits, and 2/4 cut-population runs stopped at the epoch cap. The repair that makes the windowed control transfer, per-shot standardization, is the offline form; a causal running (expanding-window/EWMA) estimator is the version that would ship and is not measured here. **Metric asymmetry.** Interpolants use full-shot neighbours while the windowed control sees two rows and the backbone the segment's past; this is deliberately adverse to the causal models but complicates direct interpretation. **Single device and model family.** Everything is one device, one diagnostic set and two closely related recurrent families; nothing here establishes transfer to another tokamak. **Uncertainty is calibrated marginally, not conditionally** (§8). **Latency** is a network-only measurement on one laptop-class machine, with up to 2× run-to-run variation from power states (§8); feature assembly and acquisition overhead are not included. **Scope.** The pedestal-top framing is inherited from the data selection (pedestal-top CES channels); no radial-dependence experiment is performed, and event-phase analyses (ELM cycle, L–H transitions) are future work—the high-variability analysis of §6.10 is an event-agnostic proxy.

11 Conclusion

We built a causal, leakage-hardened multimodal nowcaster for sparse KSTAR CES targets—a full-grid two-branch recurrent model whose V_{rot} branch never sees the fast diagnostics—and evaluated it under a pre-registered, shot-clustered, two-population protocol against a deliberately adverse bar: interpolation that sees the future, and a causal GP that is the strongest method that does not. On the observed population the model beats future-using PCHIP for ion temperature with significant, replicated skill (+0.17 to +0.32 across four independent splits and both populations), ties the strongest offline smoother, and beats every causal baseline for both targets, including the causal GP on all eight cells and including the non-adjacent regime ($\Delta t > 15$ ms).

The claim we make about deployment is now supported by both stress tests. At genuinely missing, in-domain timesteps the advantage over every causal method survives reweighting on 8/8 population–split cells; across a campaign boundary the sequence nowcaster stays ahead of PCHIP and of the causal GP on 4/4 initialisations in both populations, where the windowed control it replaced had lost its offline advantage entirely—and the difference is measured, not asserted: the fast diagnostics drift 5–11× more between campaigns than the targets, per-shot standardization repairs the control, and the sequence framing standardizes per shot and reaches the whole past by construction. An online virtual sensor competes with persistence and a causal smoother, not with an interpolant that reads the future; by that comparison the nowcaster works at the points where it matters. It emits calibrated intervals that beat both baselines' by a proper scoring rule in every cell.

We also report where it does not work and why. Rotation ties interpolation globally and its skill lives in transients; the ion-temperature win is itself a tail effect—in the median row the nowcaster equals persistence and trails future-using PCHIP by 20%, and a Huber loss pinned to the target's reproducibility bound recovers half of that bulk at no aggregate cost (conditional on the cut); the value cut that defines one of the two populations is a one-sided proxy for fit quality; the reweighting reaches 54–68% of missing T_i rows and a twentieth of missing V_{rot} rows; the predictive intervals are calibrated marginally, not per discharge; and beyond 45 ms the inclusive population is a tie. None of these is a ceiling argument, and none is left as a shrug. Each points at a specific, testable change—CES fit-quality metadata; per-window features recomputed from

the raw kHz Mirnov stream whose information 100 Hz decimation destroyed before the model ever saw it; the NBI torque channel that is absent from the dataset but is the physical cause of rotation—and the model-size axis is closed: a 21k-parameter persistence-plus-bounded-correction model matches the backbone under the cut, and a 26× width sweep is flat. So is the model-family axis: a shot-encoding neural process, a transformer pretrained on another tokamak and a 4-init ensemble all tie, and with no same-shot label no arm beats the training mean. Alongside these, the work contributes a corrected V_{rot} evaluation (54% instrument holds, removed everywhere), a two-population treatment of spectral-fit failures with the measurement that shows why neither population suffices alone, a full-grid causal framing whose reach is what clears the strongest deployable baseline, and a test-frozen selection protocol whose every decision rule was written down before the number it decided.

Reproducibility

Splits are pinned to disk and verified on reload; normalization statistics are train-file-only (plus per-shot input standardization for the sequence family); model selection never reads test data; bootstrap uses a fixed seed and 10,000 resamples; every quoted number is read from frozen evaluation artifacts by one collection script.

Code availability

The full pipeline, the pre-registration document with its per-batch execution records, the controlled-experiment runners behind every result reported here, the pinned train/validation/test splits, and the collector that regenerates every quoted number from the frozen evaluation artifacts are openly available at https://github.com/iseungsang01/ml-intern-revision. The backbone is `ces_prediction/experiments/seq/model_seq_v2.py`, the interpretable rung `model_seq_b3.py`, and the windowed control `ces_prediction/model_iter009.py` (guarded by a content hash in the test suite), so the code that produced these numbers cannot drift from the code that is released.

Data availability

The KSTAR diagnostic data underlying this study — 641 discharges, shots 30801–32751 — are not redistributed with the code. They are experimental data of the KSTAR device and are governed by the operating institution's data policy; access requests should be directed there. The repository contains everything needed to reproduce the analysis once the shot files are in place: the exact file-level split manifests, the environment configuration for every reported run, and the per-sample squared-error archives from which the bootstrap statistics are computed.

References

[1] W.H. Ko, Hyungho Lee, D. C. Seo, and Myeun Kwon. Charge exchange spectroscopy system calibration for ion temperature measurement in KSTAR. *Review of Scientific Instruments*, 81:10D740, 2010. doi: 10.1063/1.3496991.

[2] Roderick J. A. Little and Donald B. Rubin. *Statistical Analysis with Missing Data*. Wiley Series in Probability and Statistics. Wiley, 2nd edition, 2002. doi: 10.1002/9781119013563.

[3] F. N. Fritsch and R. E. Carlson. Monotone piecewise cubic interpolation. *SIAM Journal on Numerical Analysis*, 17(2):238–246, 1980. doi: 10.1137/0717021.

[4] C. E. Rasmussen and C. K. I. Williams. *Gaussian Processes for Machine Learning*. MIT Press, 2006. ISBN 0-262-18253-X.

[5] M. A. Chilenski, M. Greenwald, Y. Marzouk, N. T. Howard, A. E. White, J. E. Rice, and J. R. Walk. Improved profile fitting and quantification of uncertainty in experimental measurements of impurity transport coefficients using Gaussian process regression. *Nuclear Fusion*, 55(2):023012, 2015. doi: 10.1088/0029-5515/55/2/023012.

[6] C. Michoski, T. A. Oliver, D. R. Hatch, A. Diallo, M. Kotschenreuther, D. Eldon, M. Waller, R. Groebner, and A. O. Nelson. A Gaussian process guide for signal regression in magnetic fusion. *Nuclear Fusion*, 64 (3):035001, 2024. doi: 10.1088/1741-4326/ad1af5.

[7] Minseok Kim, W.H. Ko, Sehyun Kwak, Semin Joung, Wonjun Lee, B. Kim, D. Kim, J.H. Lee, Choongki Sung, Yong-Su Na, and Y.-C. Ghim. Kinetic profile inference with outlier detection using support vector machine regression and Gaussian process regression. *Nuclear Fusion*, 64:106052, 2024. doi: 10.1088/1741-4326/ad7304.

[8] R. L. Dougherty, A. S. Edelman, and J. M. Hyman. Nonnegativity-, monotonicity-, or convexity-preserving cubic and quintic Hermite interpolation. *Mathematics of Computation*, 52(186):471–494, 1989. doi: 10.1090/S0025-5718-1989-0962209-1.

[9] C. H. Reinsch. Smoothing by spline functions. *Numerische Mathematik*, 10(3):177–183, 1967. doi: 10.1007/BF02162161.

[10] G. Kvernadze. Convergence and Gibbs' phenomenon in cubic spline interpolation of discontinuous functions. *Journal of Computational and Applied Mathematics*, 92(2):173–188, 1998. doi: 10.1016/S0377-0427(97)00199-4.

[11] S. Eyheramendy, F. Elorrieta, and W. Palma. An irregular discrete time series model to identify residuals with autocorrelation in astronomical light curves. *Monthly Notices of the Royal Astronomical Society*, 481(4):4311–4322, 2018. doi: 10.1093/mnras/sty2487.

[12] F. Elorrieta, S. Eyheramendy, and W. Palma. Discrete-time autoregressive model for unequally spaced time-series observations. *Astronomy & Astrophysics*, 627:A120, 2019. doi: 10.1051/0004-6361/201935560.

[13] W. H. Press, S. A. Teukolsky, W. T. Vetterling, and B. P. Flannery. *Numerical Recipes: The Art of Scientific Computing*. Cambridge University Press, 3rd edition, 2007.

[14] A. H. Murphy. Skill scores based on the mean square error and their relationships to the correlation coefficient. *Monthly Weather Review*, 116(12):2417–2424, 1988. doi: 10.1175/1520-0493(1988)116<2417:SSBOTM>2.0.CO;2.

[15] Chris Bishop, C.M. Roach, and M. G. von Hellermann. Automatic analysis of JET charge exchange spectra using neural networks. *Plasma Physics and Controlled Fusion*, 35:765–773, 1993. doi: 10.1088/0741-3335/35/6/010.

[16] J. Svensson, M. von Hellermann, and R. König. Analysis of JET charge exchange spectra using neural networks. *Plasma Physics and Controlled Fusion*, 41:315–338, 1999. doi: 10.1088/0741-3335/41/2/016.

[17] Baoyue Chai, Yingying Li, Ze Chen, Wei Tao, Yixuan Zhou, Shifeng Mao, Zhengping Luo, Yi Yu, Bo Lyu, and Minyou Ye. Fast estimation of ion temperature from EAST charge exchange recombination spectroscopy using neural network. *Plasma Science and Technology*, 21:105103, 2019. doi: 10.1088/2058-6272/ab2674.

[18] J.K. Lee, W.H. Ko, H.H. Lee, B. Kim, Y.H. Lee, G.W. Shin, J. Kim, and J.M. Lee. Neural network-based analysis of charge exchange spectra in KSTAR. *Fusion Engineering and Design*, 222:115518, 2026. doi: 10.1016/j.fusengdes.2025.115518.

[19] A. Pavone, J. Svensson, A. Langenberg, U. Höfel, S. Kwak, N. Pablant, and R. C. Wolf. Neural network approximation of Bayesian models for the inference of ion and electron temperature profiles at W7-X. *Plasma Physics and Controlled Fusion*, 61:075012, 2019. doi: 10.1088/1361-6587/ab1d26.

[20] Mark D. Boyer and Jason D. Chadwick. Prediction of electron density and pressure profile shapes on NSTX-U using neural networks. *Nuclear Fusion*, 61:046024, 2021. doi: 10.1088/1741-4326/abe08b.

[21] Zichao Lin, Hongming Zhang, Fudi Wang, Cheonho Bae, Jia Fu, Yongcai Shen, Shuyu Dai, Yifei Jin, Dian Lu, Shengyu Fu, Huajian Ji, and Bo Lyu. Prediction of plasma rotation velocity and ion temperature profiles in EAST Tokamak using artificial neural network models. *Nuclear Fusion*, 64:106061, 2024. doi: 10.1088/1741-4326/ad73e8.

[22] Francisco A. Matos, Diogo R. Ferreira, and P. Carvalho. Deep learning for plasma tomography using the bolometer system at JET. *Fusion Engineering and Design*, 114:18–25, 2016. doi: 10.1016/j.fusengdes.2016.11.006.

[23] Cong Wang, Weizhe Yang, Haiping Wang, Renjie Yang, Jing Li, Zhijun Wang, Yixiong Wei, Xianli Huang, Chenshu Hu, Zhaoyang Liu, Xinyao Yu, Changqing Zou, and Zhifeng Zhao. Physics-informed deep learning model for line-integral diagnostics across fusion devices. *Nuclear Fusion*, 65:076022, 2025. doi: 10.1088/1741-4326/ade0ce.

[24] Xishuo Wei, Shuying Sun, W. M. Tang, Zhihong Lin, Hongfei Du, and Ge Dong. Reconstruction of tokamak plasma safety factor profile using deep learning. *Nuclear Fusion*, 63:086020, 2023. doi: 10.1088/1741-4326/acdf00.

[25] G.H. Zheng, Zongyu Yang, Songfen Liu, Rui Ma, Xinwen Gong, Ao Wang, Shuo Wang, and W.L. Zhong. Real-time equilibrium reconstruction by multi-task learning neural network based on HL-3 tokamak. *Nuclear Fusion*, 64:126041, 2024. doi: 10.1088/1741-4326/ad8014.

[26] S. Joung, J. Kim, S. Kwak, J. Bak, S. Lee, H. Han, H. Kim, G. Lee, D. Kwon, and Y. Ghim. Deep neural network Grad-Shafranov solver constrained with measured magnetic signals. *Nuclear Fusion*, 2020.

[27] Semin Joung, Young chul Ghim, Jaewook Kim, S. Kwak, Daeho Kwon, Chih-Jen Sung, Dong Ju Kim, Hyun-Seok Kim, J.G. Bak, and S.W. Yoon. GS-DeepNet: mastering tokamak plasma equilibria with deep neural networks and the Grad–Shafranov equation. *Scientific Reports*, 13:15799, 2023. doi: 10.1038/s41598-023-42991-5.

[28] L. Jung, Jaewook Kim, and Jisung Kang. Machine learning-based plasma profile reconstruction in KSTAR. *Journal of the Korean Physical Society*, 88:1079 – 1087, 2026. doi: 10.1007/s40042-026-01580-1.

[29] Joseph Abbate, Rory Conlin, and Egemen Kolemen. Data-driven profile prediction for DIII-D. *Nuclear Fusion*, 61:046027, 2021. doi: 10.1088/1741-4326/abe08d.

[30] Ricardo Shousha, Jaemin Seo, Keith Erickson, Z. Xing, S.K. Kim, Joseph Abbate, and Egemen Kolemen. Machine learning-based real-time kinetic profile reconstruction in DIII-D. *Nuclear Fusion*, 64:026006, 2024. doi: 10.1088/1741-4326/ad142f.

[31] Semin Joung, Jaewook Kim, S. Kwak, Kyeo reh Park, S.H. Hahn, Hyunsun Han, Hyun-Seok Kim, J.G. Bak, S. G. Lee, and Young chul Ghim. Imputation of faulty magnetic sensors with coupled Bayesian and Gaussian processes to reconstruct the magnetic equilibrium in real time. *Review of Scientific Instruments*, 89:10K106, 2018. doi: 10.1063/1.5038938.

[32] Debashish Mohapatra, Bidyadhar Subudhi, and Raju Daniel. Real-time sensor fault detection in Tokamak using different machine learning algorithms. *Fusion Engineering and Design*, 151:111401, 2019. doi: 10.1016/j.fusengdes.2019.111401.

[33] S. Kaldas, R. Rossi, I. Wyss, M. Gelfusa, and P. Gaudio. Developing autoencoder models for multi-diagnostic electron density processing in tokamaks within a scientific machine learning framework. *Journal of Fusion Energy*, 45, 2026. doi: 10.1007/s10894-026-00568-w.

[34] M.R. Wang, Chenguang Wan, Jingjing Lu, Zhi Yu, Bingjia Xiao, Yanlong Li, X.X. He, Zhengping Luo, Qiping Yuan, Yemin Hu, and Jiangang Li. Time series extrinsic regression for reconstructing missing electron temperature in tokamak. *Nuclear Fusion*, 65:076008, 2025. doi: 10.1088/1741-4326/addb5f.

[35] Zongyu Yang, Zhenghao Yang, Wenjing Tian, Jiyuan Li, Xiang Sun, Guohui Zheng, Songfen Liu, Niannian Wu, Rongpeng Li, Zhaohe Xu, Bo Li, Zhongbing Shi, Zhe Gao, Wei Chen, Xiaoquan Ji, Min Xu, and Wulyu Zhong. FusionMAE, a self-supervised pretrained model to optimize and simplify diagnostic and control of fusion plasma. *Communications Physics*, 2026. doi: 10.1038/s42005-026-02626-3.

[36] Azarakhsh Jalalvand, S.K. Kim, Jaemin Seo, Qiming Hu, M. Curie, Peter Steiner, A. Nelson, Yong-Su Na, and Egemen Kolemen. Multimodal super-resolution: discovering hidden physics and its application to fusion plasmas. *Nature Communications*, 16:8506, 2025. doi: 10.1038/s41467-025-63492-1.

[37] M. Imríšek, J. Seidl, Michal Odložilík, M. Šos, P. Bílková, Petr Boehm, Haluk Sait Aksoy, Raul Scapini Weiand, Iván Alcaraz Aguado, Robin Mahieu, and D. Naydenkova. Temporal super-resolution of kinetic plasma profiles at COMPASS using neural networks. *Plasma Physics and Controlled Fusion*, 68:065049, 2026. doi: 10.1088/1361-6587/ae72c6.

[38] HyeongJun Noh, Chweeho Heo, Xiaotian Gao, and Yong-Su Na. PanoMHD: Multimodal modelling of plasma dynamics towards Tokamak control. 2026.

[39] Zhengping Che, Sanjay Purushotham, Kyunghyun Cho, David Sontag, and Yan Liu. Recurrent neural networks for multivariate time series with missing values. *Scientific Reports*, 8:6085, 2018. doi: 10.1038/s41598-018-24271-9.

[40] Satya Narayan Shukla and Benjamin M. Marlin. Multi-time attention networks for irregularly sampled time series. 2021:14897–14911, 2021. doi: 10.48550/arxiv.2101.10318.

[41] Yulia Rubanova, Ricky T. Q. Chen, and David Duvenaud. Latent ordinary differential equations for irregularly-sampled time series. 32:5320–5330, 2019.

[42] Rich Caruana. Multitask learning. *Machine Learning*, 28:41–75, 1997. doi: 10.1023/a:1007379606734.

[43] Jiquan Ngiam, Aditya Khosla, Mingyu Kim, Juhan Nam, Honglak Lee, and Andrew Y. Ng. Multimodal deep learning. In *Proceedings of the 28th International Conference on Machine Learning (ICML)*, 2011.

[44] Tadas Baltrušaitis, Chaitanya Ahuja, and Louis-Philippe Morency. Multimodal machine learning: A survey and taxonomy. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 41:423–443, 2018. doi: 10.1109/tpami.2018.2798607.

[45] Ruibin Xiong, Yunchang Yang, Di He, Kai Zheng, Shuxin Zheng, Xing Chen, Huishuai Zhang, Yanyan Lan, Liwei Wang, and Tie-Yan Liu. On layer normalization in the transformer architecture. In *Proceedings of the 37th International Conference on Machine Learning (ICML)*, 2020.

[46] Maximilian Ilse, Jakub M. Tomczak, and Max Welling. Attention-based deep multiple instance learning. 2018. doi: 10.48550/arxiv.1802.04712.

[47] Xingjian Shi, Zhourong Chen, Hao Wang, Dit-Yan Yeung, Wai Kin Wong, and Wang-chun Woo. Convolutional LSTM network: A machine learning approach for precipitation nowcasting. 2015. doi: 10.48550/arxiv.1506.04214.

[48] Suman Ravuri, Karel Lenc, Matthew Willson, Dmitry Kangin, Rémi Lam, Piotr Mirowski, M. A. Fitzsimons, Maria Athanassiadou, Sheleem Kashem, Sam Madge, Rachel Prudden, Amol Mandhane, Aidan Clark, Andrew Brock, Karen Simonyan, Raia Hadsell, N. Robinson, Ellen Clancy, Alberto Arribas, and Shakir Mohamed. Skilful precipitation nowcasting using deep generative models of radar. *Nature*, 597:672–677, 2021. doi: 10.1038/s41586-021-03854-z.

[49] Y. U. Nam, S. Zoletnik, M. Lampert, and Á. Kovácsik. Analysis of edge density fluctuation measured by trial KSTAR beam emission spectroscopy system. *Review of Scientific Instruments*, 83:10D531, 2012. doi: 10.1063/1.4739078.

[50] G.S. Yun, W. Lee, M. Choi, Joon Bum Kim, HK Park, C. W. Domier, Benjamin Tobias, T. Liang, Xiangyu Kong, N. C. Luhmann, and A. J. H. Donné. Development of KSTAR ECE imaging system for measurement of temperature fluctuations and edge density fluctuations. *Review of Scientific Instruments*, 81:10D930, 2010. doi: 10.1063/1.3483209.

[51] J. E. Rice, A. Ince-Cushman, J.S. deGrassie, L.-G. Eriksson, Y. Sakamoto, A. Scarabosio, A. Bortolon, K.H. Burrell, B.P. Duval, C. Fenzi-Bonizec, M. Greenwald, R. J. Groebner, G. T. Hoang, Y. Koide, E. S. Marmar, A. Pochelon, and Y. Podpaly. Inter-machine comparison of intrinsic toroidal rotation in tokamaks. *Nuclear Fusion*, 47:1618–1624, 2007. doi: 10.1088/0029-5515/47/11/025.

[52] Ian Char, Youngseog Chung, Joseph Abbate, Egemen Kolemen, and Jeff Schneider. Full shot predictions for the DIII-D tokamak via deep recurrent networks. arXiv:2404.12416, 2024.